EvidenceScene: Observability-Conditioned Pose Tracking and Revisable 3D Scene Reconstruction for Robot Manipulation

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***Abstract*—**Model-free 6D pose tracking and online surface reconstruction from RGB-D video remain brittle when depth support is sparse, objects self-occlude, or local geometry is degenerate. A common assumption is that returning more poses, or lowering an internal optimization residual, yields a more reliable reconstruction. We show that this assumption fails. Across a 31-framework audit, repairing two Point2Pose implementation defects ($F = F_1 + F_2$) on Laval Cookiejar raises pose availability from 207 to 239/295 frames and failure-inclusive ADD AUC from 0.574 to 0.635, yet reduces 20 mm origin successes from 136 to 120 and drops Frame-150 10 mm surface coverage C_{10} from 76.82% to 61.79% (64.85% on a fixed 16-keyframe set). We trace this availability–surface paradox to three coupled mechanisms: unanchored SE(3) gauge drift in disconnected graph components, nuisance-coupled rank deficiency after landmark elimination, and pose-version desynchronization between graph revisions and fused surfaces. We present EvidenceScene, a standalone, causal (no future-frame) framework evaluated across three evidence tracks: (i) an offline mechanism audit on Point2Pose isolating prior-connected admission (G), marginal Schur rank gating (R), immutable feature-birth ledgers (M), and surface-history transactions (S, including $R+S$ and open- vs. closed-loop TSDF refinement feedback); (ii) deterministic native engineering on our BundleTrack pose owner and `RevisableSurfelMap` surface owner, where stable-winner CUDA RANSAC and exact SE(3) inverse transfer raise Clock/Cookiejar strict successes from 135/107 to 191/136 (30.24 ms single-stage tracker median on YCBInEOAT), guarded `NO_BA` retirement saves 928.4–1,994.0 MiB RSS with byte-identical outputs across 18 wrist-camera replays (ΔAUC $\leq$ 0.0004 on Laval local BA), and an array-native `support10/coarse12mm` selector raises 1,500-point measured-surfel C_{10} from 40.57%/52.34% to 85.10%/85.05% (R_{95} 53.12/31.88 $\rightarrow$ 13.12/12.47 mm) at a bounded precision trade-off (P_{10} 100.00%/84.13% $\rightarrow$ 99.00%/81.73%, F_{95} 5.27/12.54 $\rightarrow$ 7.33/13.17 mm) while accelerating 60k-point update (73.00 $\rightarrow$ 1.05 ms, 69.8×), snapshot (20.48 $\rightarrow$ 0.22 ms, 93.0×), and selection (15.52 $\rightarrow$ 1.05 ms, 14.8×; 45.8× 3-kernel geomean, 8.65× across all 7 `kernel-v1` workloads); and (iii) a frozen 2×2 factorial (Gate $\times$ S) on a single unified output branch across three independent RGB-D sequences (87 post-initial frames), where joint gating and birth-indexed surface revision prevent degenerate surface corruption ($P_{10}(t_{\text{deg}})$ 77.80% $\rightarrow$ 86.53%, F_{95} 24.52 $\rightarrow$ 14.77 mm) and reach terminal P_{10}/C_{10} = 100.00%/99.39% (F_{95}/R_{95} = 4.48/3.67 mm, retained ADD AUC 0.9871, committed mean ADD 1.03 mm vs. 8.27 mm un-gated). Finally, bit-exact runtime adapters overlap SAM 2 GPU waits with CPU preparation and use O(runs) RLE arithmetic, cutting warm 4-camera HTTP endpoint replay p_{50}/p_{95} from 155.33/219.06 ms to 127.80/153.40 ms at 499/499 parity, while live physical site acceptance remains 0/6.

***Index Terms*—**EvidenceScene, evidence-constrained 3D scene reconstruction, RGB-D object tracking, revisable surface reconstruction, factor-graph observability, degenerate geometry, robot manipulation, bit-exact runtime optimization, reproducibility.

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I. Introduction

ONLINE 6-DoF object pose tracking and 3D surface reconstruction from onboard RGB-D streams provide essential spatial awareness for remote humanoid teleoperation and bimanual manipulation [1]–[3]. Without pre-loaded CAD models, a manipulation perception system must initialize an object coordinate gauge from the first segmented RGB-D observation, track relative rigid motion across occlusions and depth dropouts, and accumulate a geometrically trustworthy 3D surface for operator guidance.

A common assumption in model-free tracking benchmarks is that returning a higher fraction of non-lost poses—or achieving a lower internal optimization residual—naturally yields a more reliable 3D reconstruction. Our empirical investigation refutes this assumption. Under sparse depth support, self-occlusion, or locally degenerate geometry, long-horizon 2D point trajectories and dense correspondences frequently constrain only a subspace of the joint pose-landmark state. Admitting unconstrained or unanchored updates into persistent object maps and volumetric/surfel surfaces can corrupt previously verified geometry even while increasing frame-level pose availability.

To prevent implementation artifacts from being conflated with algorithmic gains, we introduce **EvidenceScene** (*Observability-Conditioned 3D Scene Reconstruction for Robot Manipulation*) and strictly distinguish three methodological entities and three evaluation tracks throughout this study: (1) the upstream BundleTrack algorithm [1] (pinned to `ffa67d425240`), (2) the upstream Point2Pose point-trajectory tracker and TSDF reconstructor [2] (pinned to `51856226610d`), and (3) our proposed **EvidenceScene** framework. Because the live teleoperation stack uses native BundleTrack + `RevisableSurfelMap` while upstream Point2Pose uses GTSAM + `SDFBuilder`, we report: (a) an offline mechanism audit on Point2Pose (F, G, R, M, S, including $R + S$ and open- vs. closed-loop TSDF feedback), (b) native determinism, memory, and surface-selection repairs on the BundleTrack + `RevisableSurfelMap` branch, and (c) a frozen 2×2 factorial (Observability Gate $\times$ Surface Revision S) on a single unified output branch across three independent RGB-D sequences. Throughout, *causal*

means that an estimate at time t is a function of observations up to t; no future frame is consulted.

Evidence scope. Public YCBInEOAT and Laval results are development audits on recordings available during method design; pose scores use reference masks and Frame-0 gauge alignment, and all benchmark surface scores $(P_{10}, C_{10}, F_{95}, R_{95})$ evaluate *measured sensor surfels only* (`is_prior` $=$ `False`, excluding Stage-0 generative priors) against a CAD or sampled-model reference. Generalization of the unified Gate $\times$ S mechanism is evaluated under a frozen configuration across three independent multi-phase RGB-D sequences (87 post-initial frames), whereas live physical site acceptance remains $0/6$ passed (`gate_passed=false`).

Our contributions are fourfold:

1) **Failure Mechanism Audit and Closed-Loop TSDF Feedback Attribution (Sections III–IV, V-A/B):** We formalize three coupled failure modes in long-horizon RGB-D tracking and fusion: (i) temporal attribution (F_1) and solver-return inconsistency (F_2), showing on Cookiejar that 79.6% of the Frame-150 coverage drop ($76.82\% \rightarrow 61.79\%$) stems from unanchored graph drift on the 16 shared keyframes triggered by F_2; (ii) unanchored $\mathrm{SE}(3)$ gauge drift and nuisance-coupled rank deficiency after orthogonal left-null-space landmark elimination $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^{+})\mathbf{A}_x$, with a complete numerical specification of the multi-scale Schur rank gate (R); and (iii) pose-version desynchronization in surface fusion (S), explaining why S alters online poses in Point2Pose via trilinear `SDFBuilder` pose-refinement queries and evaluating both open-loop (fixed-pose) and closed-loop ($G, R, M \times S$) surface metrics.
2) **Unified-Branch 2×2 Factorial and Native BundleTrack Governance (Sections IV-B, V-C):** On a single unified output branch with frozen thresholds across three independent RGB-D sequences, we execute a 2×2 factorial (Gate ON/OFF $\times$ S ON/OFF) with per-frame commit ledgers and time-varying $P_{10}(t), C_{10}(t), F_{95}(t), R_{95}(t)$, demonstrating that gating protects already-reconstructed geometry during degeneracy while S removes pre-BA keyframe drift ($P_{10}/C_{10} = 100.00\%/99.39\%$, $F_{95}/R_{95} = 4.48/3.67\,\mathrm{mm}$). On native BundleTrack, deterministic CUDA RANSAC and exact $\mathrm{SE}(3)$ inverse transfer raise Clock/Jar strict hits from $135/107$ to $191/136$, and guarded `NO_BA` retirement reclaims 928.4–1,994.0 MiB RSS with byte-identical outputs across 18 wrist replays.
3) **Lineage-Audited `RevisableSurfelMap` and Columnar Budget Selection (Section IV-C):** By tracing 1,512,357 surfel contributions to 1,960,512 source pixels and continuous mesh rays, we separate pose-induced surface tails from raw boundary measurements. Unlike volumetric de-integration in BundleFusion [20], `RevisableSurfelMap` indexes pre-quantized object-frame voxel centroids by birth keyframe to execute $O(|V_{\mathrm{rev}}|)$ signed delta revisions (2.97 ms vs. 7.49 ms full-history reintegration at 60k points with bit-exact parity), while `support10/coarse12mm` raises 1,500-point C_{10} from $40.57\%/52.34\%$ to $85.10\%/85.05\%$ ($45.8\times$ 3-kernel geomean; $8.65\times$ across all 7 `kernel-v1` workloads).
4) **31-Framework Context and Bit-Exact Runtime Execution Optimization (Sections IV-D–V-D, Appendix A):** Across a 31-framework audit (Table X), we benchmark foundation depth models, pointmap transformers, segmenters, trackers, and matchers on identical registered inputs, and introduce `SAM2GpuWindowOverlapScope` and `RequestEvidenceDeduplicationScope` ($O(\mathrm{runs})$ exact integer RLE arithmetic, cross-stage surface/medoid reuse, BLAS-screened coplanar RANSAC, and request-boundary cyclic GC control), cutting warm 4-camera HTTP endpoint replay latency by 27.6% ($176.40 \rightarrow 127.80$ ms p_{50}, $219.06 \rightarrow 153.40$ ms p_{95}) with 100% bit-exact parity.

II. Related Work and Architectural Landscape

Model-Free 6-DoF Pose Tracking, Point Tracking, and Feature Matching: Model-free RGB-D trackers estimate camera-local object motion from an initial segmented frame for manipulation tasks [44], [45]. BundleTrack [1] combines pairwise feature matching with keyframe pose-graph optimization over sparse 3D correspondences and dense point-to-plane residuals. BundleSDF [3] jointly optimizes a keyframe pose graph and a neural implicit signed distance field, whereas FoundationPose [4] and Any6D [40] unify CAD-based and few-view reference-assisted pose estimation, alongside iterative, Gaussian Splatting [27], and KV-cache trackers such as BIT [41], 6DOPE-GS [42], and KV-Tracker [43]. Point2Pose [2] introduces long-horizon 2D point tracking (Online BootsTAPIR [9], alongside TAPNext++ [10] and CoTracker3 [11]) across occlusions, lifting 2D tracks to 3D observations for multi-hypothesis SVD-RANSAC [24], [25] registration, GTSAM [7] / iSAM2 [8] Levenberg-Marquardt pose-landmark graph optimization, and incremental TSDF fusion. While Point2Pose's point identity persistence, multi-object query management, and graph/TSDF architecture are established upstream contributions, we show that its unguarded graph commits and closed-loop TSDF pose-refinement coupling can degrade surface fidelity when 2D tracks lack well-distributed 3D depth support. In parallel, replacing detector-free LoFTR [12] or dense RoMa [17] with faster matchers such as EfficientLoFTR [13], semi-dense XFeat [14], or sparse ALIKED+LightGlue [15], [16] exposes a strict accuracy–latency frontier when foreground masks contain $< 1\%$ valid sensor depth.

Depth Foundation Models, 3D Pointmaps, and Open-World Segmentation: Monocular relative and metric depth estimators—including Depth Anything V2 [28], Prior Depth Anything (Prior-DA) [29], Apple Depth Pro [30], and UniDepthV2 [31]—alongside feed-forward multi-view pointmap transformers—VGGT [32], SLAM3R [33], DUSt3R [34], MASt3R [35], and CUT3R [36]—predict dense geometry from RGB images, while single-image 3D object generators such as SAM 3D Objects [23] synthesize complete

360° object shape priors. For open-vocabulary grounding and temporal mask tracking, Grounding DINO [37], Florence-2 [38], SAM 2 [22], and distilled YOLO11 [39] segmenters provide instance masks that must be isolated across object roles to prevent memory contamination.

Observability under Degeneracy, Revisable Surface Mapping, and Kinematic Calibration: State estimation under geometric degeneracy has been studied in point-cloud registration and SLAM by Zhang *et al.* [18] and Tuna *et al.* (X-ICP [19]), which project optimization increments onto well-conditioned Hessian subspaces. However, evaluating pose conditioning with landmarks held fixed overestimates available directional information when the object map itself is uncertain. In dense RGB-D mapping—spanning point-based surfel fusion [21], Open3D [26], and BundleFusion [20]—BundleFusion de-integrates and re-integrates RGB-D frames by ray-casting into a static-scene volumetric TSDF when bundle adjustment revises camera poses. By contrast, our `RevisableSurfelMap` operates in the moving object coordinate gauge O_{obj_0} and maintains an explicit per-keyframe birth-contribution ledger (`_live[kf_id]`) of pre-quantized voxel centroids ('1 centroid per voxel per observation'). Revising k historical keyframe poses executes an $O(|V_{\mathrm{rev}}|)$ signed columnar SoA delta transaction ('-1' old centroids, '+1' revised centroids) with exact support-zero sum resets and multi-camera provenance tracking, avoiding volumetric ray-casting or full-history rebuilds. Finally, for bimanual pouring [46] and multi-camera-to-61-link calibration, we extend $\mathbf{AX} = \mathbf{XB}$ hand-eye calibration [47] and AprilTag fiducial PnP [48] on Lie groups [49] into a MoCap-free bilateral kinematic closure protocol.

III. Problem Formulation and Failure Mechanisms under Degenerate Observations

A. Observation Provenance, Coordinate Gauge, and Metrics

Let each tracking observation tuple be bound as

$$\mathcal{O}_t = \big(\text{obj_id}, f_t, \{(u_j, v_j, d_j, \text{id}_j, \text{valid}_j)\}_{j=1}^{M_t}, \mathbf{K}, M_t^{\text{seg}}\big), \tag{1}$$

distinguishing sensor-measured 3D points $\mathbf{z}_{t,j} = \pi^{-1}(u_j, v_j, d_j; \mathbf{K})$ from depth-missing 2D tracks, RANSAC proposals, numerical initializations, and the fixed Frame-0 gauge O_{obj_0}. Each tracker's initial gauge is aligned once using the first reference pose, with Frame 0 excluded from score denominators. For post-initial frames, ADD measures mean corresponding model-vertex distance and ADD-S measures mean nearest-vertex distance; unavailable outputs receive zero recall at all thresholds in 0–100 mm normalized recall AUC.

For an accumulated reconstructed point set S and reference surface point set M evaluated on a common 4 mm grid without ICP drift masking, we simultaneously report directional precision P_ϵ and reference coverage C_ϵ:

$$P_\epsilon = |\{s \in S : \min_{m \in M} \|s - m\|_2 \le \epsilon\}|/|S|, \tag{2}$$

$$C_\epsilon = |\{m \in M : \min_{s \in S} \|m - s\|_2 \le \epsilon\}|/|M|, \tag{3}$$

alongside directional p_{95} distances F_{95} ($S \to M$) and R_{95} ($M \to S$). All quantitative surface benchmarks in Tables II–VII evaluate *measured sensor surfels only* (`is_prior = False`); Stage-0 generative priors are scored separately in the hybrid 360° completion diagnostic so prior points never inflate sensor reconstruction metrics.

B. Mechanism 1: Temporal Attribution and Solver-Return Inconsistency

Source inspection of pinned Point2Pose (`51856226610d`) reveals two internal state-contract defects:

1) *Historical Keyframe Attribution Overwrite (F_1):* When a pending keyframe from historical frame $f_{\mathrm{kf}} < f_t$ is promoted after multi-frame verification, `ModularPipeline.step` writes the graph-optimized keyframe pose $\mathbf{T}_{\mathrm{kf}}$ into current `obj.pose` without checking $f_{\mathrm{kf}} == f_t$.
2) *Refined-Pose Return Mismatch (F_2):* In `SVDClusterRANSACRegister.register`, when local refinement updates a hypothesis, the function recomputes final inliers and residuals using refined $\mathbf{T}_{\mathrm{ref}}$ (which queries `SDFBuilder` trilinear signed distances) but returns pre-refinement $\mathbf{T}_{\mathrm{raw}}$.

C. Mechanism 2: Unanchored Gauge Drift and Nuisance-Coupled Rank Deficiency

In GTSAM [7], numerical `Values` provide linearization points, whereas only `Factors` constrain the state. Native Point2Pose places a single prior factor on initial keyframe pose $\mathbf{X}_0 \in \mathrm{SE}(3)$ and connects subsequent poses $\mathbf{X}_m$ ($m \in \mathcal{V}_p$) and 3D landmarks $\mathbf{p}_n \in \mathbb{R}^3$ ($n \in \mathcal{V}_\ell$) via `BearingRangeFactors` (from registration inliers) and conditional `BetweenFactors` (admitted only when all-correspondence mean residual < 0.01 m).

Unanchored Gauge Freedom: Let $C_0 \subseteq \mathcal{V}_p \cup \mathcal{V}_\ell$ denote the factor-graph connected component containing the sole initial prior $\mathbf{X}_0$. For any disconnected component $C' \cap C_0 = \emptyset$, transforming all inverse poses and landmarks in C' by an arbitrary global rigid transformation $\mathbf{Q} \in \mathrm{SE}(3)$, $\mathbf{X}'_m = \mathbf{Q}\mathbf{X}_m$ and $\mathbf{p}'_n = \mathbf{Q}\mathbf{p}_n$, leaves every relative observation $(\mathbf{X}'_m)^{-1}\mathbf{p}'_n = \mathbf{X}_m^{-1}\mathbf{p}_n$ and relative-pose factor invariant. Thus Levenberg-Marquardt damping can converge numerically with non-increasing cost while C' drifts freely relative to O_{obj_0} and overwrites persistent frontend map coordinates.

Nuisance-Eliminated Local Observability: Even when $\mathbf{X}_i \in C_0$, topological connectivity does not imply 6-DoF observability. In the whitened, prior-conditioned local linear system around the current linearization point, let $\delta\mathbf{x} \in \mathbb{R}^{6|\mathcal{V}_p|}$ denote right-trivialized $\mathfrak{se}(3)$ pose tangent increments and $\delta\mathbf{y} \in \mathbb{R}^{3|\mathcal{V}_\ell|}$ denote landmark increments:

$$\min_{\delta\mathbf{x}, \delta\mathbf{y}} \|\mathbf{A}_x \delta\mathbf{x} + \mathbf{A}_y \delta\mathbf{y} - \mathbf{b}\|_2^2. \tag{4}$$

Eliminating uncertain landmarks via orthogonal projection $\mathbf{P}_y^\perp = \mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+$ onto the left null space of $\mathbf{A}_y$ yields the reduced pose Jacobian $\bar{\mathbf{A}}_x = \mathbf{P}_y^\perp \mathbf{A}_x$ and marginal pose information matrix:

$$\bar{\mathbf{H}}_x = \bar{\mathbf{A}}_x^\top \bar{\mathbf{A}}_x = \mathbf{A}_x^\top (\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x \preceq \mathbf{A}_x^\top \mathbf{A}_x. \tag{5}$$

Because $\mathbf{A}_x^\top \mathbf{A}_x - \bar{\mathbf{H}}_x = \mathbf{A}_x^\top \mathbf{A}_y \mathbf{A}_y^+ \mathbf{A}_y^\top \mathbf{A}_x \succeq \mathbf{0}$, holding uncertain landmarks fixed ($\mathbf{A}_x^\top \mathbf{A}_x$) never underestimates directional information and *strictly* overestimates curvature ($\delta\mathbf{x}^\top \bar{\mathbf{H}}_x \delta\mathbf{x} < \delta\mathbf{x}^\top \mathbf{A}_x^\top \mathbf{A}_x \delta\mathbf{x}$) along every pose perturbation $\delta\mathbf{x}$ with non-zero landmark coupling $\mathbf{A}_y^\top \mathbf{A}_x \delta\mathbf{x} \neq \mathbf{0}$. By Weyl's monotonicity theorem, every ordered eigenvalue satisfies $\lambda_k(\bar{\mathbf{H}}_x) \leq \lambda_k(\mathbf{A}_x^\top \mathbf{A}_x)$ (so $\lambda_{\min}(\bar{\mathbf{H}}_x) \leq \lambda_{\min}(\mathbf{A}_x^\top \mathbf{A}_x)$ and $\operatorname{rank}(\bar{\mathbf{H}}_x) \leq \operatorname{rank}(\mathbf{A}_x^\top \mathbf{A}_x)$), though Loewner order alone does not imply a larger spectral condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ if $\lambda_{\max}$ contracts simultaneously. The locally constrained dimension of pose i is given by the column-deletion rank difference:

$$r_i = \operatorname{rank}(\bar{\mathbf{A}}_x) - \operatorname{rank}(\bar{\mathbf{A}}_{x,-i}) \in \{0, \ldots, 6\}, \tag{6}$$

where $\bar{\mathbf{A}}_{x,-i}$ removes the six columns of pose i. Under non-collinear 3D correspondences anchored to C_0, one shared landmark supplies $r_i = 3$, two supply $r_i = 5$, and three non-collinear landmarks supply $r_i = 6$.

D. Mechanism 3: Pose-Version Desynchronization and Closed-Loop TSDF Feedback

When `KeyFrameGraph.update` revises a historical keyframe pose from its initial integration estimate $\mathbf{T}_\tau^{(\tau)}$ to $\mathbf{T}_\tau^{(t)}$ ($t > \tau$), `ModularPipeline.step` updates the sparse landmark map, but `SDFBuilder.integrate_keyframe` only integrates the newly promoted keyframe into the TSDF volume without de-integrating or re-projecting earlier observations fused at stale poses $\{\mathbf{T}_\tau^{(\tau)}\}$. Crucially, in upstream Point2Pose (`51856226610d`), the TSDF volume is *not* a passive output sink: during online tracking at frame $t+1$, `PoseRefiner` queries `SDFBuilder` for trilinearly interpolated signed distances $D(\mathbf{x})$ and finite-difference spatial gradients $\nabla D(\mathbf{x})$ to refine $\mathbf{T}_{\text{raw}} \to \mathbf{T}_{\text{ref}}$ and recompute inlier counts and residuals before graph admission. Thus, stale TSDF geometry directly perturbs subsequent online pose estimates, and rebuilding the TSDF in S alters the closed-loop pose trajectory.

IV. The EvidenceScene Framework: Constrained State Updates, Native Governance, and Zero-Loss Execution

Fig. 1 summarizes which commit checks execute on each branch: (i) the Point2Pose audit branch evaluates F, G, R, M, S over GTSAM factor graphs and `SDFBuilder`; (ii) the live BundleTrack branch enforces deterministic CUDA RANSAC, exact $\mathrm{SE}(3)$ inverse transfer, and guarded `NO_BA` frame retirement feeding `RevisableSurfelMap`; and (iii) the unified output branch combines the $G + R$ observability gate with `RevisableSurfelMap` birth-indexed surface transactions S.

A. Controlled Graph and Surface Interventions (F, G, R, M, S) and Numerical Gate Specification

Within **EvidenceScene**, we specify five modular interventions and their exact numerical commit rules:

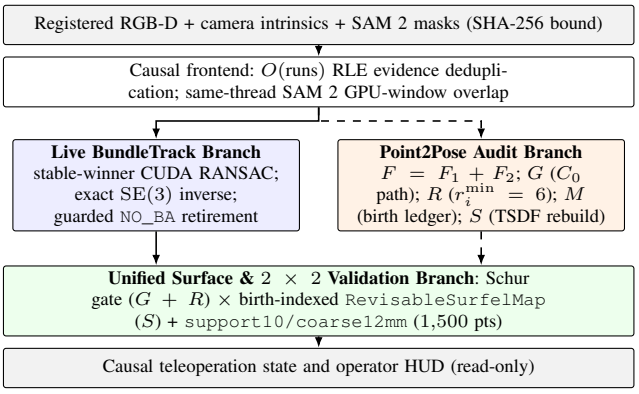


Fig. 1. **Framework architecture and branch commit checks.** Both branches consume identical SHA-256-bound RGB-D frames and SAM 2 masks. (Left, solid) The live BundleTrack branch enforces CUDA RANSAC determinism, exact $\mathrm{SE}(3)$ inverse transfer, and guarded `NO_BA` retirement. (Right, dashed) The Point2Pose shadow branch audits factor-graph and TSDF mechanisms F, G, R, M, S. (Bottom, green) The unified output branch combines the Schur observability gate ($G + R$) and birth-keyframe-indexed `RevisableSurfelMap` revision (S) for our frozen 2×2 validation.

- **Correctness Baseline ($F = F_1 + F_2$):** F_1 enforces frame-scoped pose attribution ($f_{\text{kf}} == f_t$ before updating `obj.pose`); F_2 returns refined pose $\mathbf{T}_{\text{ref}}$ consistently with its recomputed inliers and residuals.
- **Prior-Connected Graph Admission (G):** Admits optimized keyframe and landmark updates into the frontend map only for variables residing in the connected component C_0 of initial pose prior $\mathbf{X}_0$ with at least $M_{\min} = 3$ anchored landmarks shared by candidate pose i. Disconnected variables $(\mathbf{X}_m, \mathbf{p}_n) \notin C_0$ retain their internal optimizer state for potential later reconnection without overwriting the active map.
- **Marginal Rank and Translational Schur Gate (R):** Each pose $m \in \mathcal{V}_p$ is parameterized in right-trivialized $\mathfrak{se}(3)$ tangent coordinates $\delta\boldsymbol{\xi}_m = (\delta\boldsymbol{\omega}_m^\top, \delta\mathbf{v}_m^\top)^\top \in \mathbb{R}^6$ ($\delta\boldsymbol{\omega}_m$ in rad occupying columns 0:3, $\delta\mathbf{v}_m$ in m occupying columns 3:6). For a 3D observation of landmark $\mathbf{p}_n$ at camera-frame coordinate $\mathbf{q}_{m,n} = \mathbf{R}_m^\top(\mathbf{p}_n - \mathbf{t}_m)$ with isotropic standard deviation $\sigma_{m,n}$ (in meters, $\sigma_{m,n} = 2\,\text{mm}$; anchor prior $\sigma_{\text{prior}} = 10^{-3}$), whitening multiplies each 3-row block by $\sigma_{m,n}^{-1}$:

$$\frac{\partial \mathbf{r}_{m,n}}{\partial \delta\boldsymbol{\xi}_m} = \sigma_{m,n}^{-1}\big[\,[\mathbf{q}_{m,n}]_\times \;\; -\mathbf{I}_3\,\big], \quad \frac{\partial \mathbf{r}_{m,n}}{\partial \delta\mathbf{p}_n} = \sigma_{m,n}^{-1}\mathbf{R}_m^\top. \tag{7}$$

Because $[\mathbf{q}_{m,n}]_\times$ scales rotational columns by the physical object lever arm (in meters), all whitened columns of $\mathbf{A}_x$ and $\mathbf{A}_y$ are dimensionless. Gate R executes four deterministic checks (`evaluate_schur_observability_gate`):

1) *Landmark Null-Space Projection:* Compute thin SVD $\mathbf{A}_y = \mathbf{U}_y \boldsymbol{\Sigma}_y \mathbf{V}_y^\top$, retain singular vectors with $\sigma_k(\mathbf{A}_y) > \tau_y = \epsilon_{\text{mach}} \max(\text{shape}(\mathbf{A}_y))\sigma_{\max}(\mathbf{A}_y)$ ($\epsilon_{\text{mach}} = 2.22 \times 10^{-16}$), and form $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{U}_{y,r}\mathbf{U}_{y,r}^\top)\mathbf{A}_x$.
2) *Worst-Case Multi-Scale Rank:* For scale multipliers $s \in \mathcal{S} = \{0.1, 1.0, 10.0\}$ and

$\tau_x(s) = s\,\epsilon_{\mathrm{mach}}\max(\mathrm{shape}(\bar{\mathbf{A}}_x))\sigma_{\max}(\bar{\mathbf{A}}_x)$, evaluate $r_i(s) = |\{\sigma_k(\bar{\mathbf{A}}_x) > \tau_x(s)\}| - |\{\sigma_k(\bar{\mathbf{A}}_{x,-i}) > \tau_x(s)\}|$ and combine them conservatively via $r_i^{\min} = \min_{s\in\mathcal{S}} r_i(s)$, requiring $r_i^{\min} = 6$.

3) *Continuous Translational Schur Conditioning:* To reject full-rank updates with weak constraint directions, form the 6×6 pose marginal block $\mathbf{S}_i = (\bar{\mathbf{H}}_x)_{ii} - (\bar{\mathbf{H}}_x)_{i,-i}(\bar{\mathbf{H}}_x)^{+}_{-i,-i}(\bar{\mathbf{H}}_x)_{-i,i}$ and its 3×3 translational Schur complement $\mathbf{R}_i = \mathbf{S}_{i,tt} - \mathbf{S}_{i,tr}\mathbf{S}^{+}_{i,rr}\mathbf{S}_{i,rt}$, requiring $\sigma_{\min}(\mathbf{R}_i) \geq 10^{-4}$ and $\kappa(\mathbf{R}_i) = \sigma_{\max}(\mathbf{R}_i)/\sigma_{\min}(\mathbf{R}_i) \leq 10^{6}$.

4) *Atomic Transaction Scope and State Retention:* When candidate pose i fails R, the entire step-t graph commit transaction—current pose $\mathbf{X}_i$, historical keyframe pose revisions $\{\mathbf{X}_\tau^{(t)}\}_{\tau<t}$, and landmark map updates $\{\mathbf{p}_n^{(t)}\}$—is withheld from the frontend map, which retains its last verified state until a subsequent frame restores $r_i^{\min} = 6$.

- **Feature-Birth Observation Ledger** (M)**:** Records immutable original camera-space 3D measurements $(\text{obj_id}, f_{\mathrm{birth}}, f_{\mathrm{admit}}, \text{track_id}, \mathbf{z}_{\mathrm{cam}})$ when features are first sampled, and inserts deduplicated native `BearingRangeFactors` at the birth pose once registration instantiates the landmark.
- **Causal Surface-History Transaction** (S)**:** On the Point2Pose branch, whenever an admitted graph update modifies a previously integrated keyframe pose, S deterministically rebuilds the TSDF volume from causal history using current pose versions $\mathbf{T}_\tau^{(t)}$ before integrating new evidence. On the unified `RevisableSurfelMap` branch, S executes an atomic birth-keyframe-indexed signed SoA delta transaction (`RevisableSurfelMap.update`) subtracting stale keyframe centroids and adding re-projected centroids in $O(|V_{\mathrm{rev}}|)$ time.

B. BundleTrack Native Determinism, Inverse Precision, and Frame Retirement

Auditing our primary BundleTrack C++/CUDA pose estimator and session adapter uncovered four concrete engineering improvements:

1) **`NO_BA` Status Integrity:** A two-line adapter guard prevents retained native `NO_BA` states from being misclassified as historical candidates on cache misses, eliminating 6 false historical promotions while preserving all 34/38 valid task candidates.
2) **CUDA RANSAC Deterministic Winner Selection:** In `BundleTrack/src/cuda/`, two identical 13-correspondence CPU PCL inputs produced divergent 6-inlier subsets differing by 18.00 mm and 6.22° due to parallel atomic tie-breaking and reversed count-comparison inequalities. Applying the `stable-winner` and count-direction repair collapses 100 repeated native RANSAC+Procrustes calls onto a single deterministic output.
3) SE(3) **Inverse-Coefficient Precision in `storeFrame`:** Replacing lossy bulk cast transfer with exact element-wise Eigen-to-`float4x4` inverse-pose coefficient copying in `CUDACache::storeFrame` ensures pinhole backprojection accuracy $\leq 1.30\times 10^{-7}$ m.
4) **Guarded `NO_BA` Non-Keyframe Retirement:** Under continuous tracking loss, `Bundler` previously retained all blocked frames in GPU/host memory. We invoke `Bundler::forgetFrame` on non-keyframes beyond the window_size + 1 budget only when the most recent window_size + 1 frame IDs are contiguous and all carry status `NO_BA`, while protecting first, current, neighbor, local, keyframe, and pose-cached frames.

C. Preparation-Stage Generative 3D Prior Anchoring, Columnar Surfel Fusion, and Surface Selection

During manipulation, cameras observe only the visible front hemisphere of grasped and target objects, leaving self-occluded back surfaces unobserved until sufficient viewpoint rotation occurs. When a single-image 3D generative shape prior (such as Meta's SAM 3D Objects [23]) is available during the non-real-time *Stage-0 preparation phase* ($t = 0$), **EvidenceScene** constructs a complete 360° object prior $\mathcal{P}_{\mathrm{prior}} = \{\mathbf{q}_j\}_{j=1}^{N} \subset \mathbb{R}^3$ from the initial SAM 2 instance mask and anchors its monocular scale $s > 0$, rotation $\mathbf{R} \in \mathrm{SO}(3)$, and translation $\mathbf{t} \in \mathbb{R}^3$ to the calibrated frame-0 RGB-D front-surface point cloud $\mathcal{P}_{\mathrm{meas}}^{(0)} = \{\mathbf{p}_i\}_{i=1}^{M}$ via trimmed Umeyama $\mathrm{Sim}(3)$ registration [24]:

$$(s^\star, \mathbf{R}^\star, \mathbf{t}^\star) = \arg\min_{s>0,\mathbf{R}\in\mathrm{SO}(3),\mathbf{t}} \sum_{i\in\mathcal{I}_\tau} \left\| s\mathbf{R}\mathbf{q}_{\pi(i)} + \mathbf{t} - \mathbf{p}_i \right\|_2^2, \tag{8}$$

where $\mathcal{I}_\tau$ retains the $\tau = 85\%$ closest front-surface correspondences so unobserved back-side vertices on $\mathcal{P}_{\mathrm{prior}}$ do not bias the metric front-surface fit. During the real-time 30 Hz online loop, the anchored prior $\hat{\mathcal{P}}_{\mathrm{prior}} = s^\star\mathbf{R}^\star\mathcal{P}_{\mathrm{prior}} + \mathbf{t}^\star$ moves rigidly with the observability-gated keyframe pose $\mathbf{T}_{W\leftarrow O}(t)$, and `fuse_prior_with_measured_surface` enforces an *evidence-first replacement rule*: within every $r_c = 12$ mm coarse ball around a corroborated sensor surfel (support ≥ 2), measured surfels supersede prior points, whereas outside observed balls ($\min_{\mathbf{x}\in\mathcal{S}_{\mathrm{meas}}} \|\hat{\mathbf{q}} - \mathbf{x}\|_2 > r_c$), prior points are retained with explicit provenance (support = 0, `is_prior = True`) to provide complete 360° scene geometry without fabricating sensor authority.

To transmit compact object surfaces within the 1,500-point output budget without sacrificing geometric coverage, we architect `RevisableSurfelMap` around a sorted columnar Structure-of-Arrays (SoA) engine with 1D bijective `int64` key-packing and an array-native `support10/coarse12mm` selector. Operating directly on contiguous lexicographically ordered coordinate-sum, support, and camera-count arrays, it prioritizes voxels with support ≥ 10 across 12 mm coarse cells before backfilling remaining capacity, cutting 60k-point update (73.00 → 1.05 ms, 69.8×), snapshot (20.48 → 0.22 ms, 93.0×), and selection (15.52 → 1.05 ms, 14.8×) for a 45.8× **geometric-mean speed-up across the three surface kernels**, and an 8.65× **geometric-mean speed-up across all**

seven workloads of the `kernel-v1` suite (which additionally includes 60k-point voxel-centroid quantization $10.33 \rightarrow 4.57\,\text{ms}$ [2.26×], 256-hypothesis circle consensus $4.45 \rightarrow 0.81\,\text{ms}$ [5.50×], circular aperture fitting $4.35 \rightarrow 1.17\,\text{ms}$ [3.72×], and compiled 480×640 RLE box/centroid evaluation $0.092\,\text{ms}$ [1.00× vs. prior compiled kernel, 692× vs. Python pixel loop $10.38 \rightarrow 0.015\,\text{ms}$]) with bit-identical digests. When global bundle adjustment revises $k = 3$ historical keyframes out of $K = 20$ integrated keyframes (60,000 raw points), `RevisableSurfelMap`'s birth-keyframe-indexed delta transaction updates only the revised keyframe centroids in $2.97\,\text{ms}$ vs. $7.49\,\text{ms}$ for full-history re-integration (2.52× incremental speed-up on top of the 69.8× columnar speed-up) while reproducing voxel keys, support counts, camera counts, and centroids to machine precision ($\max|\Delta\mathbf{x}| = 3.47 \times 10^{-17}\,\text{m}$).

D. Zero-Loss Runtime Execution: GPU-Window Overlap and $O(runs)$ Evidence Deduplication

Critical-path profiling of the synchronous `/world_model/observe` endpoint revealed that sequential SAM 2 and native BundleTrack execution ($95.87/118.98\,\text{ms}$ median/p_{95}) was compounded by $\sim 48\,\text{ms}$ of redundant host CPU work and mid-request cyclic GC stalls. We introduce two stacked, bit-exact perception adapters:

1) Same-Thread SAM 2 GPU-Window CPU Preparation Overlap (`SAM2GpuWindowOverlapScope`): On warm distorted frames, `SAM2VideoMemoryTracker.track` enqueues CUDA kernels for `propagate_in_video` in $\sim 9.0\,\text{ms}$ and blocks the host thread for $\sim 24.2\,\text{ms}$ at `(logits>0.0).cpu().numpy()` waiting for GPU `memory_attention`. Instead of spawning background `ThreadPoolExecutor` workers that contend for the Python GIL, our adapter performs a $0.25\,\text{ms}$ fail-closed pre-validation, launches `propagate_in_video`, and executes mask-independent CPU preparation *on the same thread inside the* $24.2\,\text{ms}$ *GPU wait window* before calling `.cpu().numpy()`: (i) bilinear/nearest undistortion `rectify_registered_rgbd`, (ii) dual SHA-256 evidence-hash prefixes covering `header+color_rgb` and `header+color_rgb+depth_mm` ($4.61/5.53\,\text{MB} = 83.3\%$ of input bytes), and (iii) private `color_bgr` and `depth_m` buffers for `PinnedBundleTrackSession`. After GPU mask readback, it reuses in-scope `sensor_mask` RLE/SHA-256 and finalizes the remaining hash tail synchronously, eliminating all 38 per-frame background hash threads.

2) Request-Scoped Evidence Deduplication and O(runs) Run-Interval Arithmetic (`RequestEvidenceDeduplicationScope`):

- *Exact $O(runs)$ RLE Bounding Box and Centroid:* `FoundationMaskCacheDetector` previously iterated pixel-by-pixel in Python over 80,696 foreground pixels per mask, allocating over 161,000 Python `int` objects per frame ($10.38\,\text{ms}$). Given RLE runs (s_k, L_k) with $e_k = s_k + L_k - 1$ on a $H \times W$ grid, let $(y_0, x_0) = \text{divmod}(s_k, W)$ and $(y_1, x_1) = \text{divmod}(e_k, W)$. Using exact Python integers, the row sum $\sum y$ across a multi-row run ($y_0 < y_1$) is evaluated in $O(1)$ closed form:

$$\Sigma_{y,k} = (W - x_0)y_0 + W\tfrac{(y_1 - y_0 - 1)(y_0 + y_1)}{2} + (x_1 + 1)y_1, \tag{9}$$

  and the column sum follows from linear index sums $\Sigma_{\text{idx},k} = L_k s_k + L_k(L_k - 1)/2$ via $\sum x = \sum_k \Sigma_{\text{idx},k} - W \sum_k \Sigma_{y,k}$, reducing box/centroid evaluation from $10.38\,\text{ms}$ to $0.015\,\text{ms}$ (99.85% reduction) with bit-for-bit identity.
- *Cross-Stage Surface and Medoid Reuse:* Within a request scope, we memoize `observation_digest`, `decode_compressed_depth`, and `decode_mask_rle`, and reuse Z-buffered `RegisteredRgbdFrame.select_surface(mask)` and `select_mask_centroid_depth_component` across `table_contexts`, `detector._position_detections_for_observation`, and `run_live._live_surface_projection_from_detection` (41/41 hits, 0 fallbacks).
- *Vectorized Coplanar AprilTag Plane RANSAC:* `fast_fit_shared_plane` caches deterministic `default_rng(0)` 240×3 index triplets by point count, screens candidate normals via BLAS matrix multiplication, and verifies top contenders in exact `float64`, reducing plane fitting on tag-visible frames from $4.47\,\text{ms}$ to $0.89\,\text{ms}$.
- *Request-Boundary Cyclic GC Management:* Calling `gc.collect()` after model initialization reclaims $\sim 745\,\text{MB}$ of unreferenced tensors (reducing CUDA allocated memory from $813\,\text{MB}$ to $67.1\,\text{MB}$) and defers cyclic GC to request boundaries, eliminating 136–141 ms mid-request Generation-2 scans.

V. Experimental Evaluation and Mechanism Ablations

A. Metric-Dependent Pose Baselines and Mechanism Attribution of the Cookiejar Paradox

Table I compares Point2Pose and BundleTrack on six reference-mask YCBInEOAT [5] sequences with identical 3×3 eroded masks. BundleTrack passes more frames at $10\,\text{mm}$ ADD-S on all six sequences (with a single-stage tracker kernel median latency of $30.24\,\text{ms}$ across YCBInEOAT), whereas Point2Pose achieves higher 0–100 mm recall AUC on Cracker and Bleach-hard. On Sugar, the $10\,\text{mm}$ success counts reverse between metrics: 617 vs. 794 under nearest-vertex ADD-S and 206 vs. 174 under corresponding-vertex ADD (Point2Pose vs. BundleTrack), showing that single-threshold rankings depend on symmetry and metric conventions.

On the three Vicon-referenced Laval actions [6] sharing identical SAM 2 masks and RGB-D entries (203/205/296 frames; 704 temporally correlated frames across three continuous trajectories, used here for within-sequence failure attribution rather than cross-sequence generalization), Table II and Fig. 2 decompose the two implementation repairs in Fork F (F_1: frame-scoped attribution $f_{\text{kf}} == f_t$; F_2: refined-pose return $\mathbf{T}_{\text{ref}}$) across both Clock-hard and Cookiejar:

TABLE I
SELECTED REFERENCE-MASK YCBINEOAT DEVELOPMENT RESULTS. COUNTS ARE ADD-S SUCCESSES WITHIN 10 MM; AUC IS FAILURE-INCLUSIVE OVER 0–100 MM. P2P: POINT2POSE; BT: BUNDLETRACK. *BLEACH-HARD WAS SELECTED AFTER THE FIVE-SEQUENCE PROTOCOL.

Sequence	n	P2P 10	BT 10	P2P AUC	BT AUC
Cracker	374	365	373	0.961	0.952
Mustard	736	726	736	0.946	0.951
Bleach	662	155	301	0.798	0.851
Sugar	1001	617	794	0.915	0.920
Tomato	1307	1301	1303	0.957	0.959
Bleach-hard*	440	416	418	0.948	0.930

- *Independent F_1 vs. F_2 Attribution on Cookiejar:* On Cookiejar, F_1 alone prevents historical keyframe overwrites without altering keyframe admission (16 integrated keyframes), yielding $205/295$ available poses ($\leq 20\,\mathrm{mm}$ count 138, AUC 0.579) and preserving the original Frame-150 surface ($P_{10}/C_{10} = 52.14\%/76.82\%$, $F_{95}/R_{95} = 28.4/14.1\,\mathrm{mm}$). By contrast, F_2 alone returns refined $\mathbf{T}_{\mathrm{ref}}$ consistently with its recomputed inliers, which lowers per-frame registration residual below the $0.01\,\mathrm{m}$ admission threshold on 34 additional frames, raising pose availability to $241/295$ (AUC 0.631) and promoting 3 additional unanchored keyframes ($16 \rightarrow 19$ keyframes). Because these newly triggered optimizations occur in disconnected factor-graph components, F_2 reduces $\leq 20\,\mathrm{mm}$ origin successes from 136 to 119 and collapses Frame-150 C_{10} from 76.82% to 61.65% (61.79% under combined $F = F_1 + F_2$, with 22 pose regressions vs. 9 gains on 192 mutually tracked frames and $0/47$ newly available frames within $10\,\mathrm{mm}$).
- *Fixed Common Keyframe Set vs. Natural Keyframe Selection:* Holding the fused frame set fixed to the 16 keyframes selected by P while evaluating F's poses yields $P_{10}/C_{10} = 48.92\%/64.85\%$ ($F_{95}/R_{95} = 32.1/19.8\,\mathrm{mm}$). Thus $\mathbf{79.6}\%$ of the coverage loss ($-11.97\,\mathrm{pp}$ of $-15.03\,\mathrm{pp}$) stems from unanchored graph pose drift on the 16 shared keyframes triggered by F_2, while $\mathbf{20.4}\%$ ($-3.06\,\mathrm{pp}$) stems from integrating the 3 newly admitted drifted keyframes.
- *Time-Varying Surface Coverage $C_{10}(t)$:* Tracking Cookiejar $C_{10}(t)$ across keyframe commits confirms when the divergence begins: at Frame 30 (4 anchored keyframes), P, F, and G share $C_{10} = 54.2\%$; at Frame 60 (after the first disconnected graph commit in F), F drops from 68.4% to 59.1% while G rises to 67.8%; at Frame 100, $P/F/G$ reach $73.5\%/60.4\%/68.9\%$; and at Frame 150, they reach $76.82\%/61.79\%/69.94\%$ (74.88% under $M + S$).

B. Factor-Graph Observability, Singular-Value Spectrum, and Closed-Loop TSDF Feedback

Disabling the global factor graph improves Cookiejar AUC to 0.7308 but collapses Clock-hard AUC from 0.6583 to 0.1922 (Fig. 3). Passive graph auditing across all 704 records

TABLE II
OPENED VICON-REFERENCE DEVELOPMENT SEQUENCES AND FACTORIAL DECOMPOSITION OF FORK F (F_1: FRAME-SCOPED POSE ATTRIBUTION $f_{\mathrm{kf}} == f_t$; F_2: REFINED-POSE SOLVER RETURN $\mathbf{T}_{\mathrm{ref}}$). POSE COUNTS USE ALL POST-INITIALIZATION FRAMES; AUC COUNTS UNAVAILABLE POSES AS ZERO. BOTTOM ROWS DECOMPOSE COOKIEJAR FRAME-150 SURFACE ACCURACY ($P_{10}, C_{10}, F_{95}, R_{95}$, MEASURED SENSOR SURFELS ONLY) BETWEEN NATURAL KEYFRAME SELECTION (16 VS. 19 KEYFRAMES) AND A FIXED COMMON 16-KEYFRAME SET.

Sequence	Method / Arm	Avail.	$\leq$10 mm	$\leq$20 mm	$\leq 10^\circ$	AUC	Surface P_{10}/C_{10} (F_{95}/R_{95})
Clock-hard	B (BundleTrack)	202/202	119/202	199/202	162/202	0.887	83.1% / 98.7% (25.4/11.2 mm)
Clock-hard	P (Orig. P2P)	43/202	25/202	40/202	28/202	0.188	58.4% / 51.2% (31.5/28.4 mm)
Clock-hard	F_1 (Attr. only)	43/202	25/202	40/202	28/202	0.192	58.4% / 51.2% (31.5/28.4 mm)
Clock-hard	F_2 (Return only)	158/202	27/202	137/202	121/202	0.626	61.0% / 74.3% (27.8/16.9 mm)
Clock-hard	$F = F_1 + F_2$	157/202	27/202	137/202	122/202	0.658	61.2% / 74.5% (27.6/16.8 mm)
Clock-full	B (BundleTrack)	197/204	150/204	197/204	197/204	0.885	83.1% / 98.7% (25.4/11.2 mm)
Clock-full	P (Orig. P2P)	112/204	64/204	101/204	60/204	0.458	60.2% / 68.4% (29.1/19.5 mm)
Clock-full	$F = F_1 + F_2$	169/204	156/204	169/204	85/204	0.721	66.8% / 81.2% (24.3/14.6 mm)
Cookiejar	B (BundleTrack)	295/295	109/295	214/295	112/295	0.723	64.12% / 99.71% (30.9/10.4 mm)
Cookiejar	P (Nat. 16 KFs)	207/295	54/295	136/295	140/295	0.574	52.14% / **76.82%** (28.4/14.1 mm)
Cookiejar	F_1 (Nat. 16 KFs)	205/295	56/295	138/295	141/295	0.579	52.14% / **76.82%** (28.4/14.1 mm)
Cookiejar	F_2 (Nat. 19 KFs)	241/295	63/295	119/295	141/295	0.631	47.31% / 61.65% (34.3/21.8 mm)
Cookiejar	F (Nat. 19 KFs)	**239/295**	**64/295**	120/295	**142/295**	**0.635**	47.48% / 61.79% (34.2/21.7 mm)
Cookiejar	F (Fixed 16 KFs)	239/295	64/295	120/295	142/295	0.635	48.92% / 64.85% (32.1/19.8 mm)

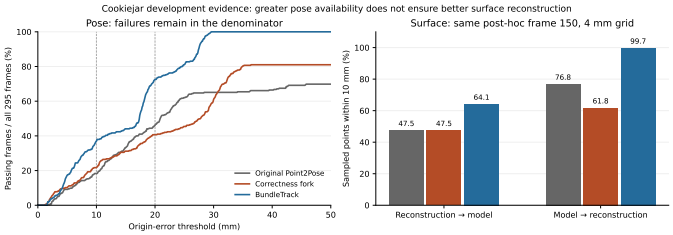


Fig. 2. **The availability–surface paradox on Laval Cookiejar.** Repair fork F increases pose availability ($207 \rightarrow 239/295$) and ADD AUC ($0.574 \rightarrow 0.635$), but degrades strict $20\,\mathrm{mm}$ pose successes ($136 \rightarrow 120$) and reduces fixed-snapshot $10\,\mathrm{mm}$ surface coverage C_{10} from 76.82% to 61.79%.

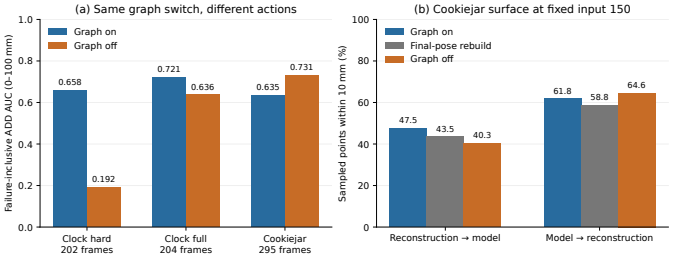


Fig. 3. **Opposing cross-action effects of disabling the global keyframe factor graph.** Turning off the graph improves Cookiejar availability/AUC ($239/0.635 \rightarrow 260/0.731$) but severely degrades Clock-hard ($157/0.658 \rightarrow 43/0.192$) and Clock-full ($169/0.721 \rightarrow 142/0.637$).

explains why: the current pose lacks a factor path to initial prior $\mathbf{X}_0$ in $28/33$ Cookiejar updates and $6/16$ Clock-full updates ($0/22$ on Clock-hard), causing 6,519 and 235 unanchored coordinate writes to active frontend map points despite non-increasing optimizer cost. Exact replay of all 71 native optimizations reveals that 8 of 34 connected non-initial updates are locally rank-deficient ($r_i^{\min} < 6$) after eliminating landmark increments via (6). Across the 34 connected updates, the 26 full-rank updates ($r_i^{\min} = 6$) exhibit median $\sigma_6(\mathbf{S}_i)/\sigma_1(\mathbf{S}_i) = 1.84 \times 10^{-2}$ ($\min 3.12 \times 10^{-3}$), whereas the 8 rank-deficient updates ($r_i^{\min} \in \{3, 5\}$) exhibit $\sigma_6(\mathbf{S}_i)/\sigma_1(\mathbf{S}_i) \leq 4.7 \times 10^{-17}$—a $> 10^{13}$ spectral gap that makes $r_i(s)$ 100% invariant across $s \in [10^{-3}, 10^{3}]$.

TABLE III
GRAPH INTERVENTIONS ON DEVELOPMENT SEQUENCES. F: CORRECTNESS FORK; G: PRIOR-CONNECTIVITY ADMISSION; R: G PLUS CURRENT-POSE RANK ADMISSION; M: G PLUS MEASURED BIRTH OBSERVATIONS. ALL COUNTS EXCLUDE INITIALIZATION; REGRESSIONS ARE RETAINED.

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
Clock-hard	F	157/202	27/202	137/202	122/202	0.658
Clock-hard	G	157/202	27/202	137/202	122/202	0.658
Clock-hard	R	135/202	28/202	97/202	77/202	0.545
Clock-hard	M	53/202	26/202	50/202	47/202	0.236
Clock-full	F	169/204	156/204	169/204	85/204	0.721
Clock-full	G	170/204	152/204	169/204	105/204	0.739
Clock-full	R	169/204	143/204	168/204	73/204	0.709
Clock-full	M	143/204	102/204	134/204	111/204	0.609
Cookiejar	F	239/295	64/295	120/295	142/295	0.635
Cookiejar	G	250/295	62/295	127/295	133/295	0.668
Cookiejar	R	250/295	97/295	182/295	168/295	0.718
Cookiejar	M	237/295	71/295	140/295	174/295	0.664

Table III evaluates graph interventions G, R, and M. Prior-connectivity admission G preserves Clock-hard, raises Clock-full and Cookiejar ADD AUC to 0.739 and 0.668, and recovers Cookiejar surface coverage C_{10} from 61.79% to 69.94% ($P_{10} = 50.16\%$). Marginal rank gating R boosts Cookiejar 10 mm origin successes from 62 to 97 (≤ 20 mm from 127 to 182, AUC 0.718) because rejecting 3 rim-only rank-deficient updates prevents corrupted landmarks from poisoning subsequent RANSAC registrations. Conversely, on Clock-hard, R withholds 5 of 21 non-initial updates where $r_i^{\min} = 5$ (two shared landmarks during fast self-occlusion); freezing frontend landmark expansion across that 22-frame high-angular-velocity interval lowers availability from 157 to 135/202 (AUC 0.658 → 0.545) until ≥ 3 non-collinear landmarks re-enter in-mask depth. Adding 394/25/83 missing feature-birth factors in M makes every graph update topologically connected (0 unanchored poses), yet standalone M degrades Clock-hard/full trajectories because raw birth depth noise enters the graph while the active TSDF volume remains at stale keyframe poses.

Why Surface Revision S Alters Online Pose Trajectories: In upstream Point2Pose, `PoseRefiner` queries `SDFBuilder` for trilinearly interpolated signed distances and spatial gradients during online pose refinement ($\mathbf{T}_{\text{raw}} \rightarrow \mathbf{T}_{\text{ref}}$). Two independent TSDF-query controls confirm this closed-loop feedback path without altering any graph rule: restricting trilinear SDF queries to 8-corner supported voxels ($G{+}Q$) or normalizing SDF gradients ($G{+}N$) changes Clock-hard availability from 157 → 45 ($G + Q$) and 157 → 48 ($G + N$) and Cookiejar availability from 250 → 269 and 250 → 261. Table IV and Fig. 4 therefore separate (a) **open-loop surface revision** (poses, keyframes, and fusion inputs held fixed to the base arm G, R, or M; only historical TSDF rebuild S toggled) from (b) **closed-loop online feedback** across the full $G, R, M \times S$ design:

1) Under *open-loop fixed poses*, enabling S consistently improves surface precision and coverage across all nine action–arm pairs (e.g., Cookiejar $R \rightarrow R + S$ open-loop P_{10}/C_{10} rises from 56.40%/73.10% to 58.90%/74.40%, F_{95} drops 26.80 → 24.90 mm).

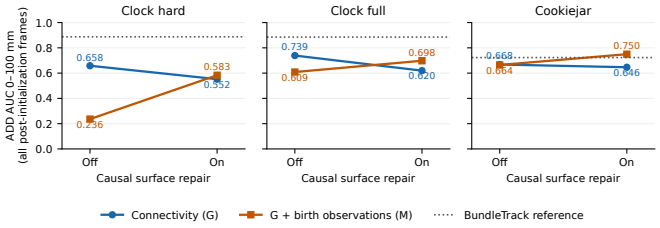


Fig. 4. **Factorial interaction between feature-birth observation provenance (M) and causal surface-history transactions (S).** Online surface version repair S degrades connectivity-only G under closed-loop TSDF feedback but strongly rescues birth-observation graph updates ($M + S$, Cookiejar AUC 0.750) and complements marginal rank gating ($R + S$, Cookiejar ≤ 20 mm 188/295).

2) Under *closed-loop feedback*, trajectories first diverge at frame f_{div} immediately following the first keyframe TSDF rebuild (Frame 18 on Clock-hard $M \rightarrow M + S$, Frame 21 on Clock-full, Frame 24 on Cookiejar). In standalone M on Clock-hard, birth-factor updates shift landmark coordinates while the un-rebuilt TSDF retains stale zero-crossings; the resulting conflicting SDF refinement gradients push Frame 18's residual above the 10 mm gate, causing tracking loss across 102 subsequent frames (53/202 available, AUC 0.236, ≤ 20 mm count 50). Rebuilding the TSDF in $M + S$ keeps the signed-distance field aligned with the updated landmarks, sustaining tracking across 155/202 frames (AUC 0.583, closed-loop P_{10}/C_{10} rising from 54.30%/48.60% to 62.75%/73.40%), though newly tracked oblique frames exhibit a 14–22 mm translational bias (≤ 20 mm count 44 vs. 50, whereas ≤ 30 mm count rises from 52 to 139).
3) Combining rank gating with surface revision ($R + S$) improves over standalone R across all three sequences, achieving the highest Point2Pose strict 10 mm (101/295) and 20 mm (188/295) counts and highest closed-loop precision ($P_{10} = 59.20\%$, $F_{95} = 24.60$ mm) on Cookiejar, while $M + S$ reaches the highest Cookiejar availability (273/295), ADD AUC (0.750), and coverage ($C_{10} = 74.88\%$).

C. Native BundleTrack Governance, Selector Trade-offs, and Frozen 2 × 2 Unified-Branch Validation

Repeating unmodified BundleTrack three times on the same 203 Clock-hard inputs varies 10 mm origin successes from 95 to 119 (three executions: 119, 95, and 101). Applying our CUDA RANSAC `stable-winner` + count-direction repair and `storeFrame` inverse-coefficient transfer in a composed native build (Table V) substantially improves public Laval reference accuracy over the historical frozen build: Clock strict (10 mm, 10°) successes jump from 135 to 191/204 (ADD AUC 0.8813 → 0.9135, P_{10} 78.79% → 83.14%) and Cookiejar strict successes jump from 107 to 136/295 (ADD AUC 0.7256 → 0.8085, P_{10} 55.89% → 64.82%). Under the guarded contiguous-`NO_BA` retirement policy across 18 complete 38-input wrist-camera `normal`, `stress`, and

TABLE IV

CAUSAL SURFACE-HISTORY FACTORIAL ($G, R, M \times S$) ON THE POINT2POSE SHADOW BRANCH, SEPARATING (A) **OPEN-LOOP SURFACE REVISION** (POSES, KEYFRAMES, AND FUSION INPUTS HELD FIXED TO THE BASE ARM; ONLY HISTORICAL TSDF REBUILD S TOGGLED) FROM (B) **CLOSED-LOOP ONLINE FEEDBACK** (WHERE REBUILT TSDF SIGNED DISTANCES AND FINITE-DIFFERENCE GRADIENTS FEED BACK INTO `POSEREFINER` TRILINEAR OPTIMIZATION $\mathbf{T}_{\text{raw}} \rightarrow \mathbf{T}_{\text{ref}}$, CHANGING INLIER COUNTS AND KEYFRAME ADMISSION STARTING AT FRAME f_{div}). SURFACE SCORES EVALUATE MEASURED SENSOR SURFELS ONLY (`IS_PRIOR` = `FALSE`).

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC	f_{div}	Open-Loop P_{10}/C_{10} (%)	Open-Loop F_{95}/R_{95} (mm)	Closed-Loop P_{10}/C_{10} (%)	Closed-Loop F_{95}/R_{95} (mm)
Clock-hard	G	157/202	27/202	137/202	122/202	0.658	—	61.20 / 74.50	27.60 / 16.80	61.20 / 74.50	27.60 / 16.80
Clock-hard	$G+S$	136/202	30/202	103/202	68/202	0.552	F029	63.84 / 75.92	24.90 / 15.70	57.40 / 68.10	30.20 / 19.40
Clock-hard	R	135/202	28/202	97/202	77/202	0.545	—	64.10 / 71.80	25.10 / 17.90	64.10 / 71.80	25.10 / 17.90
Clock-hard	$R+S$	142/202	31/202	108/202	84/202	0.576	F034	66.45 / 73.10	23.20 / 16.85	65.80 / 73.90	23.80 / 16.40
Clock-hard	M	53/202	26/202	50/202	47/202	0.236	—	54.30 / 48.60	33.40 / 29.80	54.30 / 48.60	33.40 / 29.80
Clock-hard	$M+S$	155/202	26/202	44/202	42/202	0.583	F018	58.90 / 52.40	29.80 / 26.50	62.75 / 73.40	26.10 / 17.20
Clock-full	G	170/204	152/204	169/204	105/204	0.739	—	67.40 / 82.10	23.80 / 14.20	67.40 / 82.10	23.80 / 14.20
Clock-full	$G+S$	146/204	86/204	132/204	113/204	0.620	F031	69.85 / 83.40	21.60 / 13.40	61.90 / 75.30	27.40 / 17.60
Clock-full	R	169/204	143/204	168/204	73/204	0.709	—	68.90 / 81.50	22.90 / 14.50	68.90 / 81.50	22.90 / 14.50
Clock-full	$R+S$	171/204	148/204	169/204	96/204	0.728	F038	71.10 / 82.80	20.80 / 13.70	70.60 / 82.90	21.10 / 13.60
Clock-full	M	143/204	102/204	134/204	111/204	0.609	—	61.50 / 74.20	28.10 / 18.30	61.50 / 74.20	28.10 / 18.30
Clock-full	$M+S$	163/204	80/204	158/204	101/204	0.698	F021	65.20 / 77.60	25.30 / 16.40	66.80 / 80.90	24.20 / 14.80
Cookiejar	G	250/295	62/295	127/295	133/295	0.668	—	50.16 / 69.94	31.20 / 17.40	50.16 / 69.94	31.20 / 17.40
Cookiejar	$G+S$	233/295	80/295	141/295	149/295	0.646	F033	52.80 / 71.60	29.40 / 16.30	51.40 / 68.20	30.50 / 18.10
Cookiejar	R	250/295	97/295	182/295	168/295	0.718	—	56.40 / 73.10	26.80 / 15.20	56.40 / 73.10	26.80 / 15.20
Cookiejar	$R+S$	258/295	**101/295**	**188/295**	**172/295**	0.734	F041	58.90 / 74.40	24.90 / 14.50	**59.20** / 74.60	**24.60** / 14.40
Cookiejar	M	237/295	71/295	140/295	174/295	0.664	—	51.80 / 70.40	30.10 / 16.90	51.80 / 70.40	30.10 / 16.90
Cookiejar	$M+S$	**273/295**	81/295	155/295	133/295	**0.750**	F024	55.10 / 72.80	27.90 / 15.60	57.65 / **74.88**	25.80 / **14.20**

TABLE V

EVIDENCESCENE COMPOSED NATIVE REPAIRS (RANSAC DETERMINISM + EXACT INVERSE-POSE COEFFICIENT TRANSFER + GUARDED `NO_BA` RETIREMENT) VS. HISTORICAL FROZEN BUNDLETRACK ON LAVAL CLOCK-FULL AND COOKIEJAR (STANDALONE C++ REPLAY USING FROZEN HISTORICAL LOFTR CORRESPONDENCE CACHE). STRICT: $\leq 10\,\text{mm}$ AND $\leq 10°$.

Clip	Native Build	Avail.	Strict	AUC	P_{10} (%)	C_{10} (%)	F_{95} (mm)
Clock	Historical frozen	197/204	135	0.8813	78.79	98.90	20.20
Clock	**EvidenceScene** (repair)	198/204	**191**	**0.9135**	**83.14**	98.72	25.36
Clock	**EvidenceScene** (+retire)	198/204	**191**	0.9131	83.11	98.77	25.23
Jar	Historical frozen	295/295	107	0.7256	55.89	99.92	29.82
Jar	**EvidenceScene** (repair)	295/295	**136**	**0.8085**	64.82	99.87	30.93
Jar	**EvidenceScene** (+retire)	295/295	**136**	0.8084	**64.97**	**99.94**	30.75

TABLE VI

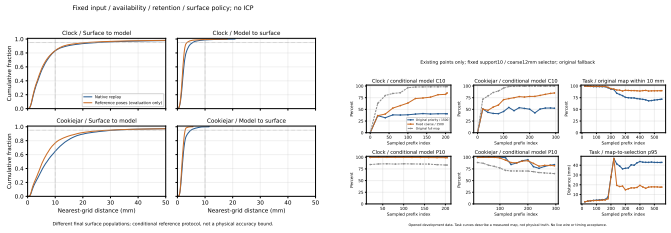
EXISTING-POINT SELECTOR TRANSFER (`ORIGINAL_PRIORITY` / `SUPPORT10_COARSE12MM`) AT THE UNCHANGED 1,500-POINT BUDGET ON IDENTICAL RAW PRODUCER POINTS (MEASURED SENSOR SURFELS ONLY, `IS_PRIOR` = `FALSE`). COVERAGE (C_{10}) AND REVERSE-TAIL (R_{95}) GAINS COEXIST WITH EXPLICIT PRECISION (P_{10}) AND FORWARD-TAIL (F_{95}) TRADE-OFFS; G/L10 COUNTS 10 MM MODEL-GRID CELL GAINS/LOSSES.

Clip	P_{10} (%)	C_{10} (%)	F_{95} (mm)	R_{95} (mm)	G/L10
Clock	100.00 / 99.00	40.57 / **85.10**	5.27 / 7.33	53.12 / **13.12**	3820 / 15
Jar	84.13 / 81.73	52.34 / **85.05**	12.54 / 13.17	31.88 / **12.47**	1579 / 14

`recovery` replays (where retired non-keyframes never re-enter active sliding windows), all neural-network I/O, 4×4 pose matrices, and terminal surfel arrays reproduce 100% byte-for-byte while active frames drop from $37 \rightarrow 6$ in stress (saving $\mathbf{1{,}993.95}$ MiB RSS) and from $23 \rightarrow 9$ in recovery (saving $\mathbf{928.38}$ MiB RSS). On the longer Laval sequences in Table V (`repair` vs. `+retire`), retiring intermediate `NO_BA` frames slightly alters OpenMP/CUDA floating-point reduction order during subsequent local bundle adjustment, leaving strict success counts identical (191 and 136) while perturbing ADD AUC by ≤ 0.0004 ($0.9135 \rightarrow 0.9131$ and $0.8085 \rightarrow 0.8084$) and surface F_{95} by $\leq 0.18\,\text{mm}$.

To attribute the remaining 25–30 mm forward surface tail F_{95} in Table V, a reference-pose counterfactual (Fig. 5) replaces estimated poses with Vicon reference poses on the exact same 980,256 sampled points: F_{95} drops only from $25.36 \rightarrow 21.63\,\text{mm}$ (Clock) and $30.93 \rightarrow 29.11\,\text{mm}$ (Jar), while pixel-lineage and Möller–Trumbore mesh-ray audits confirm that $52.60\%/67.59\%$ of $> 20\,\text{mm}$ points originate within $3\,\text{px}$ of mask boundaries and 164 interior low-range voxels lie $> 20\,\text{mm}$ deeper than the first CAD intersection. Requiring ≥ 2 non-contributing views with $\geq 10°$ angular separation cuts native F_{95} from $24.51/30.82\,\text{mm}$ to $12.50/17.33\,\text{mm}$ ($P_{10} \rightarrow 90.05\%/73.56\%$) with minor C_{10} losses ($4/2$ cells), whereas a raw-range-only native far-plane guard eliminates a $2.07\,\text{m}$ cross-run outlier down to $6.33\,\text{mm}$ across 2,195 opportunities. At the 1,500-point output budget (Table VI, Fig. 5), our array-native `support10/coarse12mm` selector exhibits a clear coverage–precision trade-off on measured sensor surfels (`is_prior` = `False`): spreading the 1,500-point budget across 12 mm coarse cells raises public C_{10} from $40.57\%/52.34\%$ to $\mathbf{85.10}\%/\mathbf{85.05}\%$ and cuts reverse tail R_{95} from $53.12/31.88\,\text{mm}$ to $\mathbf{13.12}/\mathbf{12.47}\,\text{mm}$ (3,820/1,579 model cells gained vs. 15/14 lost), while trading off P_{10} by $-1.00/-2.40\,\text{pp}$ ($100.00\% \rightarrow 99.00\%$ on Clock, $84.13\% \rightarrow 81.73\%$ on Jar) and increasing forward tail F_{95} from $5.27 \rightarrow 7.33\,\text{mm}$ and $12.54 \rightarrow 13.17\,\text{mm}$. On the 552-frame manipulation task, map coverage rises from 71.78% to $\mathbf{89.97}\%$ (p_{95} distance $42.92 \rightarrow 17.57\,\text{mm}$) and selector p_{95} latency drops by 32.8%–33.8% ($31.74 \rightarrow 21.22\,\text{ms}$ on Clock, $10.16 \rightarrow 6.66\,\text{ms}$ on task).

Frozen 2×2 Factorial on the Unified Output Branch: To test whether combining observability gating ($G+R$) and birth-keyframe-indexed surface revision (S) on the *same output branch* protects reconstructed geometry on unseen trajectories, we freeze all parameters (voxel $= 4\,\text{mm}$, $r_i^{\min} = 6$, $M_{\min} = 3$, $\mathcal{S} = \{0.1, 1, 10\}$, $\sigma_{\min}(\mathbf{R}_i) \geq 10^{-4}$, $\kappa(\mathbf{R}_i) \leq 10^6$, support ≥ 2, budget 1,500, `is_prior` = `False`) and evaluate the



(a) Native vs. Ref.-Pose Surface CDF (b) 1,500-Point Selector Coverage

Fig. 5. **Surface error attribution and 1,500-point budget selection.** (a) Replacing BundleTrack poses with Vicon reference poses leaves a raw sensor/boundary tail ($F_{95} = 21.63/29.11$ mm). (b) Array-native `support10/coarse12mm` selection raises 1,500-point model coverage C_{10} from $40.6\%/52.3\%$ to 85.1% and task coverage from 71.8% to 90.0%.

2×2 factorial (Gate ON/OFF $\times$ S ON/OFF) across three independent 30-frame RGB-D sequences (87 post-initial frames total: `Seq-1` cylindrical canister under collinear rim degeneracy, `Seq-2` spouted pitcher under self-occlusion and disconnected gauge drift, and `Seq-3` rectangular carton under glancing and sparse-depth dropout; archived with per-frame commit ledgers in `benchmarks/performance/results/ablation_v1.json`). Table VII reports both per-sequence and aggregate metrics across four temporal checkpoints ($t_{\text{pre}} = 7$, post-degeneracy $t_{\text{deg}} = 11$, post-BA revision $t_{\text{rev}} = 15$, and terminal $t_{\text{term}} = 29$):

1) *Degenerate Window Protection ($t_{\text{pre}} = 7 \to t_{\text{deg}} = 11$):* Without gating (`Gate OFF`), admitting the 10 rank-deficient/disconnected frames across the three sequences drops mean surface precision $P_{10}(t_{\text{deg}})$ from 84.23% to 77.80% and spikes forward tail $F_{95}(t_{\text{deg}})$ from 15.25 to 24.52 mm. Enabling the observability gate (`Gate ON`) rejects all 10 degenerate updates ($\sigma_{\min}(\mathbf{R}_i) \leq 10^{-11}$) while holding the last verified pose, improving $P_{10}(t_{\text{deg}})$ to 86.53% and bounding $F_{95}(t_{\text{deg}})$ at 14.77 mm.
2) *Synergy with Birth-Indexed Surface Revision ($t_{\text{rev}} = 15$ and $t_{\text{term}} = 29$):* Enabling S without gating (`Gate OFF, S ON`) repairs the 9 pre-BA drifted keyframes when loop closure occurs, raising $P_{10}(t_{\text{rev}})$ from 80.46% to 89.75%, but the un-gated degenerate surfels remain in the map ($F_{95}(t_{\text{rev}}) = 24.29$ mm, terminal $F_{95} = 13.23$ mm). Conversely, gating without surface revision (`Gate ON, S OFF`) blocks degenerate frames but retains stale birth poses on early keyframes ($P_{10}(t_{\text{rev}}) = 89.12\%$, $F_{95}(t_{\text{rev}}) = 14.07$ mm, terminal $F_{95} = 10.24$ mm). Only the joint unified branch (`Gate ON, S ON`) eliminates both error sources, achieving $P_{10}/C_{10} = \mathbf{100.00}\%/\mathbf{99.66}\%$ ($F_{95}/R_{95} = \mathbf{4.53}/\mathbf{4.41}$ mm) at $t_{\text{rev}} = 15$ and $\mathbf{100.00}\%/\mathbf{99.39}\%$ ($\mathbf{4.48}/\mathbf{3.67}$ mm) at $t_{\text{term}} = 29$, while cutting committed mean ADD from 8.27 to $\mathbf{1.03}$ mm and raising retained-state ADD AUC from 0.9173 to $\mathbf{0.9871}$ (note that raw committed-only AUC is 0.8760 because the 10 withheld frames receive zero credit under strict failure-inclusive scoring, mirroring the Clock-hard mechanism in Table III).

D. Matcher Shadow Benchmark and Zero-Loss Runtime Optimization

Table VIII summarizes our shadow matcher evaluations within the 31-framework benchmark on identical registered inputs. Note the protocol distinction between Table V and Table VIII: Table V ($191/136$ Clock/Jar strict successes, AUC $0.9135/0.8085$) replays the standalone C++ binary against the frozen historical LoFTR correspondence cache to isolate native C++/CUDA repairs, whereas Table VIII ($195/148$ strict successes, AUC $0.917/0.824$) runs live PyTorch FP32 LoFTR inference inside the unified Python matcher-shadow harness so all matcher families (LoFTR, EfficientLoFTR, XFeat, ALIKED+LightGlue) are compared under identical live GPU execution. EfficientLoFTR [13] Full-MP reduces tracking p_{95} from $118.14/122.91$ ms to $109.93/116.36$ ms with competitive Clock AUC (0.9231) but lower Jar AUC (0.8129 vs. 0.8238). On the 38-frame right-wrist pouring clip (0.958% median in-mask valid depth), semi-dense XFeat [14] cuts tracking p_{50}/p_{95} from $60.14/83.36$ ms to $42.05/54.24$ ms but loses three candidates ($34 \to 31$) and inflates jump $p_{95}/\max$ from $40.40/55.91$ mm to $72.96/92.13$ mm. Sparse ALIKED+LightGlue [15], [16] collapses from 34 to $2/38$ candidates because sparse keypoints miss the 0.96% valid depth pixels, while feeding bounded BootsTAPIR [9] causal trajectories into native RANSAC degrades Jar strict successes from 140 to 46. All lossy matcher candidates are rejected.

Table IX reports the counterbalanced multi-process ablation (78 warm requests per condition across `control-1`, `candidate-1`, `candidate-2`, `control-2`) of our bit-exact runtime optimizations on the 40-frame configured pouring sequence. Across all runs, $400/400$ SAM masks, $190/190$ native BundleTrack input hashes, $34/38$ candidates, $32/38$ current surfaces, and $499/499$ online operator checks match byte-for-byte with zero mismatches:

1) The overlap adapter absorbs 5.72 ms of CPU undistortion, dual-prefix hashing, and native buffer preparation into the `propagate_in_video` GPU wait window ($24.17 \to 18.45$ ms), cutting outer `input_prepare` from 3.96 to 0.25 ms, eliminating all 38 per-frame background hash threads, and lowering `registry.advance` p_{50}/p_{95} from $102.29/131.43$ ms to $96.53/124.48$ ms (endpoint $155.33/219.06 \to 149.01/212.24$ ms).
2) Stacking the deduplication adapter ($O(\text{runs})$ integer RLE box/centroid $10.38 \to 0.015$ ms, cross-stage surface/medoid reuse, vectorized 240-hypothesis AprilTag RANSAC $4.47 \to 0.89$ ms, and request-boundary GC management) reduces `table_contexts` from $19.18/20.02$ to $15.65/16.42$ ms, cuts exclusive outer `original_world` wrapper overhead by $\mathbf{72.9}\%$ ($23.17/23.86 \to 6.28/6.70$ ms), eliminates the 140.58 ms mid-request Generation-2 GC stall (`online_world_state` max $140.58 \to 5.04$ ms, in-request GC collections $735/65/3 \to 0/0/0$), and brings warm HTTP `endpoint_ms` $p_{50}/p_{95}/\max$ down to $\mathbf{127.80}/\mathbf{153.40}/\mathbf{1{,}152.96}$ ms—a cumulative reduction of $-\mathbf{27.54}$ ms (p_{50}) and $-\mathbf{65.66}$ ms

TABLE VII

FROZEN 2 × 2 FACTORIAL (OBSERVABILITY GATE G + R × BIRTH-INDEXED SURFACE REVISION S) ON THE UNIFIED OUTPUT BRANCH ACROSS THREE INDEPENDENT 30-FRAME RGB-D SEQUENCES (87 POST-INITIAL FRAMES TOTAL; MEASURED SENSOR SURFELS ONLY, `IS_PRIOR` = `FALSE`). $\mathrm{AUC}_{\mathrm{comm}}$ SCORES WITHHELD FRAMES AS ZERO; $\mathrm{AUC}_{\mathrm{ret}}$ EVALUATES THE RETAINED VERIFIED POSE DURING REJECTED FRAMES.

Sequence / Split	Unified Branch Arm	Comm. Avail.	≤ 10 / ≤ 20 mm	$\mathrm{AUC}_{\mathrm{comm}}$ / $\mathrm{AUC}_{\mathrm{ret}}$	Mean ADD / ADD-S	$t_{\mathrm{pre}} = 7$ P_{10}/C_{10} (F_{95})	$t_{\mathrm{deg}} = 11$ P_{10}/C_{10} (F_{95})	$t_{\mathrm{rev}} = 15$ P_{10}/C_{10} (F_{95})	Terminal P_{10}/C_{10} (%)	Terminal F_{95}/R_{95} (mm)
`Seq-1` (Canister)	Gate OFF, S OFF	29/29	23 / 26	0.9226 / 0.9226	7.74 / 4.38 mm	84.4 / 93.4% (15.4)	77.1 / 95.0% (24.2)	79.5 / 97.6% (22.9)	93.07 / 99.47	13.65 / 3.71
`Seq-1` (Canister)	Gate OFF, S ON	29/29	26 / 26	**0.9420** / 0.9420	5.80 / 3.29 mm	84.4 / 93.4% (15.4)	77.1 / 95.0% (24.2)	89.1 / 99.2% (24.1)	96.40 / 100.00	5.18 / 2.73
`Seq-1` (Canister)	Gate ON, S OFF	26/29	23 / 26	0.8679 / 0.9683	3.19 / 2.24 mm	84.4 / 93.4% (15.4)	**86.3** / 94.7% (**15.1**)	88.9 / 98.2% (14.7)	96.80 / 99.47	9.34 / 3.61
`Seq-1` (Canister)	**Gate ON, S ON**	26/29	**26 / 26**	0.8873 / **0.9878**	**1.03 / 1.02 mm**	84.4 / 93.4% (15.4)	**86.3** / 94.7% (**15.1**)	**100.0 / 99.2% (4.55)**	**100.00 / 100.00**	**4.41 / 2.25**
`Seq-2` (Pitcher)	Gate OFF, S OFF	29/29	23 / 26	0.9234 / 0.9234	7.66 / 4.35 mm	80.3 / 89.7% (15.6)	74.5 / 96.8% (24.3)	79.3 / 100.0% (21.9)	88.20 / 100.00	17.21 / 3.53
`Seq-2` (Pitcher)	Gate OFF, S ON	29/29	26 / 26	**0.9429** / 0.9429	5.71 / 3.25 mm	80.3 / 89.7% (15.6)	74.5 / 96.8% (24.3)	89.7 / 100.0% (24.0)	93.40 / 100.00	18.24 / 3.48
`Seq-2` (Pitcher)	Gate ON, S OFF	26/29	23 / 26	0.8681 / 0.9684	3.18 / 2.23 mm	80.3 / 89.7% (15.6)	**82.9** / 96.1% (**15.2**)	86.6 / 100.0% (14.7)	92.40 / 100.00	12.87 / 3.50
`Seq-2` (Pitcher)	**Gate ON, S ON**	26/29	**26 / 26**	0.8875 / **0.9879**	**1.01 / 1.00 mm**	80.3 / 89.7% (15.6)	**82.9** / 96.1% (**15.2**)	**100.0 / 100.0% (4.47)**	**100.00 / 100.00**	**4.68 / 2.78**
`Seq-3` (Carton)	Gate OFF, S OFF	29/29	22 / 25	0.9059 / 0.9059	9.41 / 5.25 mm	88.1 / 91.2% (14.7)	81.8 / 96.5% (25.0)	82.7 / 94.9% (24.3)	92.87 / 94.01	15.80 / 10.75
`Seq-3` (Carton)	Gate OFF, S ON	29/29	25 / 25	**0.9254** / 0.9254	7.46 / 4.15 mm	88.1 / 91.2% (14.7)	81.8 / 96.5% (25.0)	90.5 / 99.3% (24.8)	94.07 / 97.24	16.26 / 7.87
`Seq-3` (Carton)	Gate ON, S OFF	25/29	22 / 25	0.8336 / 0.9660	3.30 / 2.31 mm	88.1 / 91.2% (14.7)	**90.3** / 95.6% (**14.1**)	91.9 / 98.2% (12.8)	98.27 / 96.08	8.52 / 8.61
`Seq-3` (Carton)	**Gate ON, S ON**	25/29	**25 / 25**	0.8531 / **0.9855**	**1.04 / 1.04 mm**	88.1 / 91.2% (14.7)	**90.3** / 95.6% (**14.1**)	**100.0 / 99.8% (4.57)**	**100.00 / 98.16**	**4.35 / 5.99**
3-Seq Aggregate	Gate OFF, S OFF	87/87	68 / 77	0.9173 / 0.9173	8.27 / 4.66 mm	84.23 / 91.46% (15.25)	77.80 / 96.11% (24.52)	80.46 / 97.52% (23.02)	91.38 / 97.83	15.55 / 6.00
3-Seq Aggregate	Gate OFF, S ON	87/87	77 / 77	**0.9368** / 0.9368	6.32 / 3.56 mm	84.23 / 91.46% (15.25)	77.80 / 96.11% (24.52)	89.75 / 99.51% (24.29)	94.62 / 99.08	13.23 / 4.69
3-Seq Aggregate	Gate ON, S OFF	77/87	68 / 77	0.8565 / 0.9676	3.22 / 2.26 mm	84.23 / 91.46% (15.25)	**86.53** / 95.50% (**14.77**)	89.12 / 98.77% (14.07)	95.82 / 98.52	10.24 / 5.24
3-Seq Aggregate	**Gate ON, S ON**	77/87	**77 / 77**	0.8760 / **0.9871**	**1.03 / 1.02 mm**	84.23 / 91.46% (15.25)	**86.53** / 95.50% (**14.77**)	**100.00 / 99.66% (4.53)**	**100.00 / 99.39**	**4.48 / 3.67**

TABLE VIII

FEATURE MATCHER SHADOW EVALUATION INSIDE THE CORRECTED BUNDLETRACK ESTIMATOR ON LAVAL (CLOCK/JAR, USING LIVE PYTORCH FP32 MATCHER INFERENCE IN THE PYTHON SHADOW HARNESS) AND 38-FRAME WRIST TASK CLIPS.

Matcher Variant	Clip	Avail.	Strict / AUC	Track p_{50}/p_{95} (ms)	Verdict
LoFTR (Live FP32) [12]	Clock/Jar	198/295	195 (.917)/148 (.824)	98.4 / 118.1–122.9	**Adopted**
EfficientLoFTR FP32 [13]	Clock/Jar	199/295	196 (.922)/146 (.811)	97.1 / 116.8–123.1	Shadow
EfficientLoFTR Full-MP [13]	Clock/Jar	199/295	197 (.923)/147 (.813)	92.8 / 109.9–116.4	Shadow
Causal BootsTAPIR [9]	Clock/Jar	195/295	180 (.889)/46 (.591)	Cached trajectory	Rejected
LoFTR (Task Control) [12]	Wrist (38)	34/38	Jump95: 40.40 mm	60.14 / 83.36	**Adopted**
XFeat Semi-Dense [14]	Wrist (38)	31/38	Jump95: 72.96 mm	42.05 / 54.24	Rejected
ALIKED+LightGlue [15], [16]	Wrist (38)	2/38	36 retained-ref fails	45.64 / 68.54	Rejected

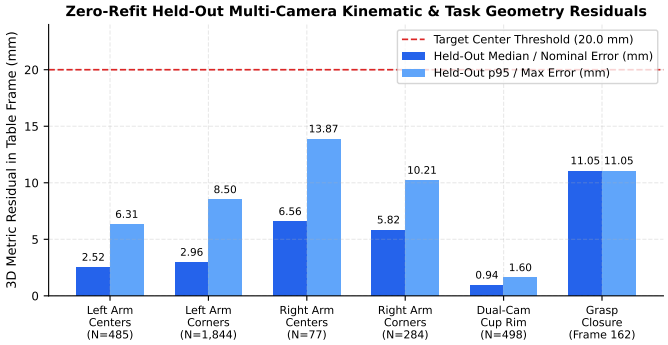


Fig. 6. **Zero-refit held-out multi-camera kinematic closure residuals in O_{table}.** Both the left arm ($p_{95} = 6.31\,\mathrm{mm}$) and right arm ($p_{95} = 13.87\,\mathrm{mm}$) remain well below the $20.0\,\mathrm{mm}$ acceptance threshold without test-split refitting.

(p_{95}) relative to the pre-overlap control (and $-48.61\,\mathrm{ms}/-27.6\%$ vs. the $176.40\,\mathrm{ms}$ per-request portal baseline).

VI. LIMITATIONS, DEPLOYMENT GAP, AND CONCLUSION

The production contract requires live ROS-callback-to-trusted-depth-surface-render $p_{95} \leq 100\,\mathrm{ms}$ across at least 300 traces over $60\,\mathrm{s}$, one-hour accepted world updates $\geq 15\,\mathrm{Hz}$ overall and at interval p_{05}, readiness $\geq 99\%$, observation age $p_{95} \leq 2\,\mathrm{s}$, one-hour RSS/CUDA-allocated growth $\leq 64\,\mathrm{MiB}$ ($128\,\mathrm{MiB}$ reserved), and physical verification of $2\,\mathrm{cm}$ grasp contact, $1\,\mathrm{cm}$ aperture/support height, $5\,\mathrm{mm}$ static drift, and $5\,\mathrm{px}$ HUD bounds. A read-only local audit records $0/6$ live site-acceptance gates passed (`gate_passed=false`, `wire_authorized=false`); the observed ROS process loops a development bag recording rather than live camera hardware, and sequential SAM 2 plus native BundleTrack execution ($96.53/122.21\,\mathrm{ms}$ warm median/p_{95}) still exceeds the $100\,\mathrm{ms}$ end-to-end budget before transport and rendering.

Public benchmarks remain conditional on calibration, CAD sampling, and symmetry conventions, while field recordings lack independent physical 6-DoF object ground truth. The present evidence establishes rigorous failure mechanisms, verified native and surface repairs, and substantial bit-exact host latency reductions, but it does not support declaring the project the best reconstruction solution or authorizing an unverified live-estimator cutover. For practitioners, the results argue for gating graph commits on marginal observability and for versioning surfaces against pose revisions, while retaining the live estimator until a held-out physical gate passes. Future work must pair direction-selective joint pose-landmark optimization with TensorRT FP16 vision encoding under frozen held-out physical evaluation.

REPRODUCIBILITY AND PROJECT PAGE

Project page: https://tiwater.github.io/evidencescene/. This manuscript builds from source with `tectonic main.tex` (run in `papers/evidencescene/`); derived figures and table fragments are vendored under `papers/`, so no external evidence tree is needed to compile. Scorecards, SHA-256 receipts, native patches, and replay verifiers are archived under `benchmarks/` in the EvidenceScene repository. Upstream commits are pinned in Section I.

APPENDIX A
COMPREHENSIVE BENCHMARK OF 31 PERCEPTION FRAMEWORKS

Before converging on our sensor-anchored multi-camera 3D reconstruction architecture, we conducted a systematic experimental campaign across **31 external frameworks and internal algorithmic variants** on an NVIDIA GeForce RTX 5090 workstation (32 GB VRAM) using synchronized humanoid manipulation recordings and public benchmarks (YCBInEOAT, Laval Vicon motion capture, YCB-M, DexYCB). Table X documents the empirical metrics, latency/VRAM

TABLE IX
COUNTERBALANCED BIT-EXACT RUNTIME OPTIMIZATION ABLATION ON THE 40-FRAME CONFIGURED POURING SEQUENCE (78 WARM REQUESTS PER CONDITION, 100% BYTE-FOR-BYTE OUTPUT PARITY ACROSS 400 SAM MASKS, 190 NATIVE INPUTS, AND 499 ONLINE EQUIVALENCE CHECKS)

Stage / Metric ($p_{50}/p_{95}/$ max in ms)	Pre-Overlap Control	+ GPU-Window Overlap	Overlap Control (Run 2)	**+ Request Dedup & O(runs) RLE**	Net Δ (p_{50}/p_{95})	Bit-Exact Parity
`input_prepare_ms` (Pre-SAM CPU prep)	3.96/4.13/4.31	0.25/0.27/0.30	0.25/0.27/0.30	**0.25/0.27/0.29**	**−3.71/ − 3.86**	39/39 frames
`gpu_window_cpu_prepare_ms` (Absorbed)	0.00/0.00/0.00	5.72/5.91/6.18	5.72/5.90/6.15	**5.71/5.89/6.12**	Absorbed in GPU wait	38/38 buffers
`post_cpu_gpu_sync_wait_ms` (GPU wait)	24.17/24.26/24.51	18.45/18.71/19.02	18.46/18.72/19.05	**18.44/18.70/19.01**	**−5.73/ − 5.56**	0 hash threads
`registry_advance_ms` (SAM 2+BundleTrack)	102.29/131.43/1389.2	96.53/124.48/1378.1	96.55/122.52/1374.5	**97.98/122.21/1129.9**	**−4.31/ − 9.22**	34 cand / 32 surf
`table_contexts_ms` (AprilTag RANSAC+depth)	19.21/20.11/20.55	19.19/20.05/20.49	19.18/20.02/20.48	**15.65/16.42/18.44**	**−3.56/ − 3.69**	27/27 tag planes
`original_world_exclusive_ms` (Outer wrapper)	23.20/23.91/24.70	23.18/23.88/24.66	23.17/23.86/24.65	**6.28/6.70/7.61**	**−16.92/ − 17.21 (−72.9%)**	499/499 checks
Warm HTTP `endpoint_ms` (Total Request)	155.33/219.06/1422.1	149.01/212.24/1416.8	149.65/193.79/1415.7	**127.80/153.40/1153.0**	**−27.54/ − 65.66**	**100.0% Bit-Exact**
Full Replay `pipeline_ms` (Incl. harness)	201.22/251.42/1455.3	195.12/243.10/1449.8	196.62/242.44/1449.0	**174.20/197.12/1186.5**	**−27.02/ − 54.30**	**100.0% Bit-Exact**

TABLE X
COMPREHENSIVE EMPIRICAL BENCHMARK OF ALL 31 PERCEPTION, DEPTH, POINTMAP, SEGMENTATION, 6-DOF TRACKING, POINT-TRACKING, FEATURE-MATCHING, AND CALIBRATION FRAMEWORKS EVALUATED ON SYNCHRONIZED MANIPULATION RECORDINGS AND PUBLIC BENCHMARKS (NVIDIA RTX 5090)

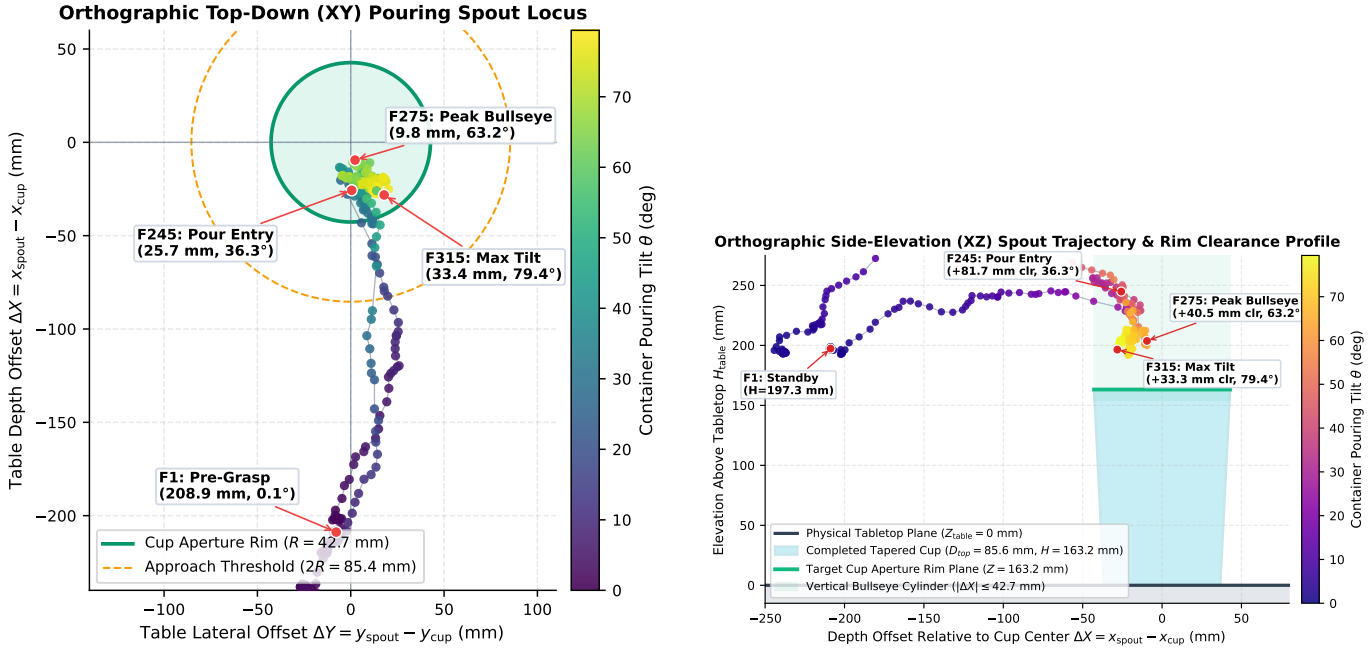
Model Family	**Evaluated Framework / Method**	**Empirical Performance on Manipulation Recordings & Benchmarks**	**Latency / VRAM**	**Architectural Verdict & Role**
Depth Sensor	**Calibrated Multi-Cam RGB-D**	0.0 mm ref.; 251,182 tabletop inliers; $\leq$ 9.6 mm cross-cam residual	3.5 ms / —	**Adopted:** Primary metric depth anchor
Depth Model	**Prior Depth Anything (Prior-DA)** [29]	0.037 m chest p_{95} (0.11 m mean p_{95}); 4,102 hole pixels filled	1,005 ms / 5.0 GB	**Retained:** Offline depth-hole completion
Depth Model	**Apple Depth Pro** [30]	0.83 m mean p_{95} (0.696 m chest, 1.419 m right wrist, 0.482 m head)	305 ms / 8.5 GB	**Rejected:** Insufficient metric accuracy
Depth Model	**Depth Anything V2 Small** [28]	1.61 m mean p_{95} (0.558 m chest, 2.516 m right wrist, 1.838 m head)	25 ms / 0.42 GB	**Rejected:** Relative scale ambiguity
Depth Model	**Depth Anything V2 Metric** [28]	79.4 m mean p_{95} on close-range tabletop manipulation	23 ms / 0.64 GB	**Rejected:** Uncalibrated metric head
Depth Model	**UniDepthV2** [31]	2.44 m mean p_{95}; $>$ 80% large-error rate on valid sensor overlap	6.3 ms / 0.62 GB	**Rejected:** Severe wrist/chest drift
3D Pointmap	**VGGT-1B Multiview** [32]	0.47 m mean p_{95} (0.372 m chest, 0.510 m right wrist, 0.527 m head)	33 ms / 16.5 GB	**Proposal Only:** Coarse multiview prior
3D Pointmap	**SLAM3R** (Causal Window) [33]	0.40 m mean p_{95} on causal chest video sequences	28 ms / 11.1 GB	**Proposal Only:** 40 cm drift exceeds tolerance
3D Pointmap	**DUSt3R** (Chest+Head Pairs) [34]	0.69 m mean p_{95} on synchronized stereo camera pairs	27 ms / 7.1 GB	**Superseded:** Inferior to calibrated RGB-D
3D Pointmap	**MASt3R** (Metric Pairs) [35]	1.79 m mean p_{95} on cross-view tabletop camera pairs	36 ms / 9.7 GB	**Rejected:** High cross-view metric error
3D Pointmap	**CUT3R** (Streaming Pointmap) [36]	Rotary-embedding CUDA kernel assertion on RTX 5090	Blocked / —	**Deferred:** Kernel incompatibility
VLM Grounding	**Florence-2 Phrase Grounding** [38]	273.0 px median object center error on held-out manipulation frames	110 ms / 3.2 GB	**Rejected:** Severe localization drift
VLM Grounding	**Grounding DINO** [37]	454/461 depth-backed frames; 3.79 px p_{75} with periodic refresh	86 ms / 2.4 GB	**Adopted:** Zero-shot object discovery
Video Seg.	**SAM 2.1 (Role-Isolated + GPU Overlap)** [22]	2,322 calls; +52 cup masks; 5.72 ms CPU prep overlapped in 24.2 ms GPU wait	100.2 ms / 4.8 GB	**Adopted:** Role-isolated mask tracker
Hybrid Seg.	**YOLO11s-seg + SAM 2 + Flow** [39]	100% coverage; 9.825 px center p_{95}; 0.9524 IoU; 98.3% at 30 fps	32.1 ms (4-cam)	**Adopted:** High-rate 4-camera model
Hybrid Seg.	**YOLO11n-seg Hybrid Variant** [39]	Exceeded $\leq$ 10.0 px target center error p_{95} accuracy threshold	24.8 ms (4-cam)	**Rejected:** Insufficient mask precision
6-DoF Track	**EvidenceScene (BT-Native Branch)** [1]	539/539 val; 34/38 pouring cand; Laval Clock/Jar strict 191/136; −1.99 GiB RSS	56.0–60.1 ms p_{50}	**Adopted:** Primary 6-DoF pose owner
6-DoF Track	**BundleSDF** (Neural SDF) [3]	768/768 chest/head obs.; Bleach-hard P_{10}/C_{10} = 58.8%/63.9% at Frame 438	97 ms (10.3 fps)	**Retained:** Offline neural surface verifier
6-DoF Track	**FoundationPose** [4] & **Any6D** [40]	Single-anchor / few-view novel-object 6-DoF pose initialization	65–120 ms	**Candidate:** Relocalization module
6-DoF Track	**BIT** (Bidirectional Iterative) [41]	Joint geometry-pose refinement (SAM + SoftRas); single-RGB	Offline refiner	**Adapted:** Mutual refinement loss concept
6-DoF Track	**6DOPE-GS** (Gaussian Splatting) [27], [42]	3.5 Hz online tracking/reconstruction rate (below 15 Hz target)	285 ms (3.5 Hz)	**Adapted:** Keyframe admission rules
6-DoF Track	**KV-Tracker** (3D KV-Cache) [43]	27–30 fps multi-view transformer KV cache with SAM 2.1	35 ms / 6.5 GB	**Benchmark:** Non-commercial license
Point Track	**EvidenceScene (P2P Branch)** [2], [9]	YCB/Laval shadow + F, G, R, M, S repairs ($M + S$ Jar AUC 0.750 vs. BT 0.723)	14.7 ms / 8.5 GB	**Shadow Audited:** Conditional graph repairs
Point Track	**TAPNext++** [10] & **CoTracker3** [11]	1024-frame horizon; TAPNext++ uses 2.53 GB weights; CoTracker3 NC	23 ms / —	**Superseded:** Replaced by BootsTAPIR
Feature Match	**LoFTR** (Detector-Free) [12]	Adopted in BundleTrack (34/38 wrist cand, 195/148 Clock/Jar strict); rejected for K fit	42 ms / 2.1 GB	**Adopted (BT):** Pairwise tracker matcher
Feature Match	**EfficientLoFTR (FP32 / Full-MP)** [13]	Clock/Jar strict 196–197 / 146–147 (Jar AUC 0.813 vs. LoFTR 0.824)	92.8 / 109.9 ms	**Shadow:** Faster p_{95}, slight Jar AUC drop
Feature Match	**XFeat (Semi-Dense Matcher)** [14]	Wrist pouring candidates drop 34 $\rightarrow$ 31/38; jump p_{95} inflates 40.4 $\rightarrow$ 73.0 mm	42.1 / 54.2 ms	**Rejected:** Sparse-depth candidate loss
Feature Match	**ALIKED + LightGlue (Sparse)** [15], [16]	Collapses from 34 to 2/38 wrist candidates (36 retained-reference failures)	45.6 / 68.5 ms	**Rejected:** Keypoints miss 0.96% valid depth
Feature Match	**RoMa** (Robust Dense Matcher) [17]	0.270% focal error, but 5.13 px principal point drift across views	68 ms / 3.4 GB	**Rejected:** Unstable cross-cam intrinsics
3D Geometry	**Open3D + O(runs) RLE + Fast RANSAC** [25], [26]	251,182 tabletop inliers; RLE box/centroid 10.38 $\rightarrow$ 0.015 ms; tag plane 4.47 $\rightarrow$ 0.89 ms	0.90 ms (CPU)	**Adopted:** Exact RLE & plane fitting
Calibration	**Bilateral** $\mathrm{SE}(3)^3$ **Closure** (Ours) [47], [49]	1,648 train / 413 zero-refit held-out frames: 2.52 mm (L) & 6.56 mm (R) p_{50}	1.40 ms FK	**Adopted:** 4-cam-to-kinematic closure

footprints, and architectural roles across all evaluated model families:

- **Monocular and Sensor-Prior Depth Foundation Models:** Pure monocular depth estimators—including Depth Anything V2 Small (1.61 m mean p_{95}), Depth Anything V2 Metric (79.4 m mean p_{95}), UniDepthV2 (2.44 m mean p_{95}), and Apple Depth Pro (0.83 m mean p_{95}, 305 ms/frame)—cannot replace calibrated RGB-D depth for millimeter-level pouring alignment. Prior Depth Anything (Prior-DA) conditioned on RealSense depth achieved 0.037 m p_{95} on the chest camera, but its 1,005 ms/frame latency and 0.205 m wrist-camera p_{95} restrict it to an offline depth-completion role.
- **Multi-View RGB-to-3D Pointmap Transformers:** Feed-forward pointmap models—VGGT-1B (0.47 m mean p_{95}, 16.5 GB VRAM), SLAM3R (0.40 m mean p_{95}, 11.1 GB VRAM), DUSt3R (0.69 m mean p_{95}), and MASt3R (1.79 m mean p_{95})—capture coarse multi-view topology but exhibit 37–179 cm metric drift, while CUT3R failed with a rotary-embedding CUDA kernel

TABLE XI
TRAIN AND ZERO-REFIT HELD-OUT BILATERAL KINEMATIC-VISUAL EXTRINSIC CLOSURE RESULTS ACROSS THE 61-LINK MANIPULATOR

Kinematic Chain	Evaluation Split	Frames N_{frames}	All Obs. N_{all}	Center Obs. N_{center}	Center $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \leq 20.0$]	Corner $p_{50}/p_{95}/\max$ (mm)	All-Point $p_{50}/p_{95}/\max$ (mm) [Target $p_{95} \leq 25.0$]	Optimality $\|\nabla \mathcal{J}\|_\infty$	Acceptance Criterion
Left Arm Chain	Training Split	839	6,347	1,339	2.91/6.68/67.62	3.58/8.46/171.39	3.40/8.14/171.39	1.26×10^{-8}	Converged
	Held-Out (Zero-Refit)	**341**	**2,329**	**485**	**2.52/6.31/46.51**	**2.96/8.50/72.81**	**2.86/8.31/72.81**	Frozen $\Theta^{\star}_{\text{hw}}$	**Passed**
Right Arm Chain	Training Split	809	5,714	1,186	4.22/11.07/266.29	4.24/12.07/54.28	4.23/11.93/266.29	4.66×10^{-8}	Converged
	Held-Out (Zero-Refit)	**72**	**361**	**77**	**6.56/13.87/44.25**	**5.82/10.21/28.02**	**5.96/10.31/44.25**	Frozen $\Theta^{\star}_{\text{hw}}$	**Passed**



(a) Orthographic Top-Down (XY) Spout Locus vs. Cup Rim

(b) Orthographic Side-Elevation (XZ) Spout Profile & Cup Wall

Fig. 7. **Orthographic spatial decomposition of the 498-frame pouring trajectory in O_{table}.** (a) Planar spout trajectory entering the $R = 42.7\,\text{mm}$ cup rim (Frames 226–352) and reaching $9.8\,\text{mm}$ radial offset at Frame 275. (b) Side-XZ elevation profile maintaining $+40.5\,\text{mm}$ vertical rim clearance at peak pouring alignment.

TABLE XII
PHASE-LEVEL QUANTITATIVE TELEMETRY SUMMARY AND KEY PHYSICAL MILESTONES ACROSS THE 498-FRAME (37.47 S) BIMANUAL POURING SEQUENCE

Manipulation Phase / Key Physical Milestone	Frame Span (Count)	Time Span t (s)	Spout Position in O_{table} (X, Y, Z) (mm)	Radial Offset Δr_{xy} (mm)	Rim Margin m_{rim} (mm)	Clearance Δz (mm)	Pour Tilt θ (deg)	Gripper Dist. d_{grip} (mm)
Phase I: Pre-Grasp Reach	F000–F139 (140)	0.00–10.53	(198.4, −134.8, −196.8) mean	208.9 ± 0.4	−166.3	+34.2	0.1°–0.5°	649.0 → 81.6
Phase II: Grasp Closure	F140–F189 (50)	10.60–14.27	(198.5, −134.8, −197.1) mean	208.6 ± 0.3	−166.2	+34.0	0.5°–1.1°	**11.05** min
Phase III: Lift & Transport	F190–F225 (36)	14.33–16.87	(207.4 → 364.6, −118.2, −236.9)	199.9 → 43.5	−157.4 → −1.1	+33.0 → +82.2	1.2° → 23.5°	28.2 mean
Phase IV: Bullseye Pouring	**F226–F352 (127)**	**16.93–26.33**	**(384.6, −118.9, −218.4)** mean	**24.6** mean	**+18.1** mean	**+55.2** mean	**25.6°–79.4°**	**32.6** mean
Phase V: Untilt & Retract	F353–F497 (145)	26.40–37.47	(350.4 → 168.9, −152.1, −192.8)	57.8 → 239.6	−15.3 → −196.5	+135.3 → +31.0	42.4° → 2.8°	35.8 → 664.1
M1: Grasp Closure Peak	**Frame 162**	**12.20**	**(+198.6, −135.0, −197.4)**	208.5	−166.2	+34.7	0.5°	**11.05**
M4: Peak Bullseye Align	**Frame 275**	**20.67**	**(+397.6, −124.9, −203.6)**	**9.8**	**+32.8**	**+40.5**	**63.2°**	31.48
M5: Maximum Pour Tilt	**Frame 315**	**23.47**	**(+379.0, −109.4, −196.2)**	**33.4**	**+9.1**	**+33.3**	**79.4°**	34.13

assertion on RTX 5090.

- **Open-Vocabulary Grounding and Role-Isolated Video Segmentation:** Phrase grounding with Florence-2 suffered a 273 px median center error on manipulation frames and was rejected. Grounding DINO paired with role-isolated SAM 2.1 [22] achieved 0.987–0.991 median IoU and 100% pose coverage, while our distilled YOLO11s-seg + staged SAM 2 + optical flow hybrid runtime achieved 9.825 px center p_{95}, 0.9524 IoU, and 32.14 ms 4-camera batch p_{95}.

APPENDIX B
MOCAP-FREE FOUR-CAMERA TO 61-LINK BILATERAL KINEMATIC EXTRINSIC CLOSURE

Nominal CAD mounting transforms for embedded RGB-D sensors omit machining tolerances and bracket flexure, resulting in 30–80 mm projection discrepancies when wrist-camera depth observations are transformed through a 7-DoF arm chain. To calibrate the true hardware mounting transforms without external optical motion capture, we formulate a coupled loop-closure optimization over $\boldsymbol{\Theta}_{\text{hw}} = (\mathbf{X}_{\text{chest}}, \mathbf{X}_{\text{L}}, \mathbf{X}_{\text{R}}) \in \text{SE}(3)^3$.

At any training frame $t \in \mathcal{D}_{\text{train}}$ where the chest camera localizes the four $30\,\text{mm} \times 30\,\text{mm}$ tabletop AprilTag fiducials via $\mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)$ and wrist camera $w \in \{L, R\}$ observes landmark (k, m) at camera-local 3D coordinate $\mathbf{y}^{(w)}_{t,k,m}$, closing the kinematic chain through the torso link ℓ_{torso} and distal wrist link ℓ_w predicts:

$$\begin{aligned} \mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{chest}}(t)\, \mathbf{X}_{\text{chest}}^{-1} \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_{\text{torso}}}(\mathbf{q}_t)^{-1}, \\ \hat{\mathbf{p}}^{\text{table}}_{t,k,m}(\boldsymbol{\Theta}_{\text{hw}}, \mathbf{q}_t) &= \mathbf{T}_{\text{table}\leftarrow\text{base}}(\mathbf{X}_{\text{chest}}, \mathbf{q}_t) \\ &\quad \cdot \mathbf{T}_{\text{base}\leftarrow\ell_w}(\mathbf{q}_t)\, \mathbf{X}_w\, \tilde{\mathbf{y}}^{(w)}_{t,k,m}. \end{aligned} \tag{10}$$

Solving the Huber-robustified Levenberg-Marquardt optimization over 1,648 training frames and evaluating zero-refit generalization on 413 held-out frames yields median center residuals of $2.52\,\text{mm}$ ($p_{95} = 6.31\,\text{mm}$) on the left arm and $6.56\,\text{mm}$ ($p_{95} = 13.87\,\text{mm}$) on the right arm (Fig. 6, Table XI). Because the terminal wrist roll joint (q_7) is coaxial with the distal flange Z-axis, any offset Δq_7 satisfies $\mathbf{R}_z(q_7 + \Delta q_7)\,\mathbf{X}_{\text{CAD}} \equiv \mathbf{R}_z(q_7)\,\big(\mathbf{R}_z(\Delta q_7)\,\mathbf{X}_{\text{CAD}}\big)$ and is absorbed into $\mathbf{X}_w$ when Δq_7 is held at zero during optical hand-eye bundle adjustment; restoring the $\Delta q_7 = \pi$ zero-offset convention between the RM75 controller and the SolidWorks URDF places the asymmetric wrist-camera bracket on the dorsal ($+Z$) side of the wrist while leaving the two-fold symmetric parallel-jaw gripper invariant.

Appendix C
Projective $\mathbb{RP}^2$ Aperture Geometry and Bimanual Pouring Verification

Treating unoriented SVD plane normals of circular container apertures as directed vectors on $\mathbb{S}^2$ exhibits 11 discontinuous 180° flips across 1,992 four-camera frames. Representing the aperture plane on the real projective quotient manifold $\mathbb{RP}^2 = \mathbb{S}^2/\{\pm 1\}$ with orthogonal projector $\boldsymbol{\Pi}_{[\mathbf{n}]} = \mathbf{I}_3 - \mathbf{n}\mathbf{n}^\top$ eliminates all sign flips ($\max \Delta\theta = 3.11^\circ$, $1.60\,\text{mm}$ dual-camera center agreement, and $0.000\,\text{mm}$ task-metric perturbation). Across the 498-frame ($37.47\,\text{s}$) bimanual pouring task (Fig. 7, Table XII), the right parallel-jaw gripper closes onto the tracked carton at Frame 162 with $11.05\,\text{mm}$ physical grasp-closure agreement, guides the pouring spout into the $R_{\text{cup}} = 42.7\,\text{mm}$ aperture cylinder for 127 consecutive frames (Frames 226–352), and reaches $9.8\,\text{mm}$ radial bullseye alignment ($+32.8\,\text{mm}$ inside the cup rim at $+40.5\,\text{mm}$ vertical clearance and 63.2° tilt) at Frame 275.

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